

The Long Lasting Effects of Warfare on Tax Progressivity: Evidence from World War II*

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June 10, 2026

Abstract

This paper studies how World War II reshaped progressive taxation. Treating the war as a natural experiment in a panel of 35 countries from 1929 to 2010, we estimate how much active participation raised a country's top marginal income tax rate relative to what it would have been without the war. Our central finding is that the war raised top rates sharply and durably: five years after entry the top rate is about 38 percentage points higher, and three decades out about 42 points higher; the effect stays large for three decades and remains statistically distinct from zero for about five, before fading as top rates fall across countries after 1980. The estimates are robust to a wide range of inference methods and specifications, and the same design applied to World War I yields a similar but smaller effect. We interpret the persistence as a shift between political-economic equilibria rather than a change in preferences: by compressing the wealth distribution, the war induced a new equilibrium that kept top taxation high.

*We thank Martin Eichenbaum, Giorgio Primiceri, Matthew Rognlie, and Matthias Doepke for their guidance throughout this project, and Joel Mokyr for insights that were central to it. We are grateful to participants in the Macro Lunch seminar series at Northwestern University for helpful feedback. All errors are our own.

1 Introduction

Why did top marginal income tax rates rise so steeply in some countries in the mid-twentieth century, and why did those increases last as long as they did? In the countries that mobilized for World War II, top rates climbed to historic highs in the 1940s and stayed elevated for several decades, narrowing toward other countries only much later (Scheve and Stasavage, 2016). Both the size of the rise and its timing point to the war itself. We argue that this wartime shock accounts for much of the divergence in how progressively these countries have taxed ever since.

The main empirical finding of this paper is that World War II raised top marginal tax rates sharply and durably in the countries that mobilized for it. Treating the war as a natural experiment, we estimate its effect on top rates in a panel of 35 countries from 1929 to 2010.¹ Five years after entry a war-active country's top rate is about 38 percentage points higher than the no-war path; ten years out, about 40; three decades out, about 42. The effect stays large for roughly three decades and remains statistically distinct from zero for about five, before fading as top rates fall across countries after 1980. These estimates are robust to a wide range of inference methods and specifications.

Two forces connect mass-mobilization warfare to higher top rates: the fiscal demands of financing it and the political mobilization of the citizens called on to fight. The first is fiscal: waging total war required revenue on a scale pre-war tax systems could not supply, and the progressive income tax was the tool best able to raise it quickly from those most able to pay (Harrison, ed, 1998). The second is political: as states conscripted ordinary citizens to fight, democratic publics demanded that owners of wealth bear a comparable burden, a conscription of wealth to match the conscription of labor (Scheve and Stasavage, 2010).

Beyond the top income rate, two further measures point toward higher progressivity in the war-active countries. The top statutory rate on inheritances, a separate levy aimed at large fortunes, rises after war entry; and the redistribution wedge — the gap between pre-tax and post-tax inequality — widens, indicating that the tax-and-transfer system bore more heavily on the top. Section 4 takes up both, and together they indicate that progressivity rose beyond the top income rate alone.

¹Each estimate is cumulative: it measures how much higher a war-active country's top rate is, a given number of years after it enters the war, than it would have been without the war, with the no-war path identified from the non-war-active countries. Because top rates are persistent, the effect at a given horizon compounds the initial increase rather than capturing only that year's change; Section 4 defines this cumulative object precisely.

The war sharply widened the gap in top tax rates between war-active and non-war-active countries, and the increase persisted for decades. The raw data show a gap that is fairly constant before the war, on the order of fifteen to twenty percentage points, then widens sharply in the 1940s, stays wide for decades, and narrows only as rates fall in both groups after 1980 (Figure 1). Section 4 shows that this widening is the war’s effect.

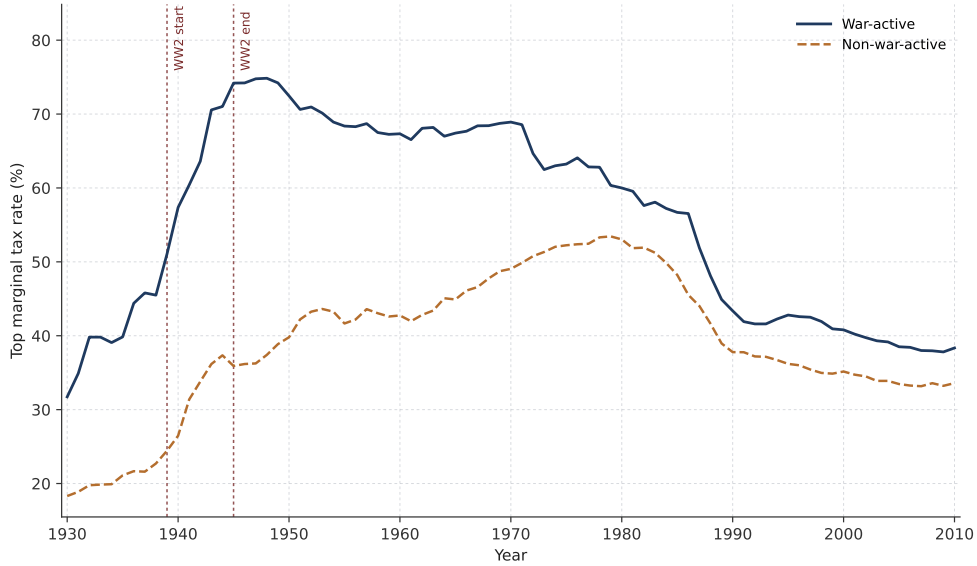


Figure 1: Average top marginal income tax rate by year, 1930–2010, for the 13 war-active³ and 22 non-war-active countries in the main sample. Each line is the unweighted cross-country mean of the top statutory rate within the group. Vertical lines mark the start and end of the war.

We explain the persistence of the high top rates as a shift to a new political-economic equilibrium rather than as a by-product of a larger postwar state. The leading alternative is the ratchet effect: war creates permanent revenue needs, so the fiscal state stays large (Higgs, 1987). But a larger state explains the level of postwar revenue, not why the burden kept falling on top incomes, and the persistence of the top rate does not track the growth of government. In the account we develop, the war compressed the wealth distribution; the more equal distribution shifted political choices toward more progressive taxation; and that taxation in turn held the distribution compressed—a self-reinforcing loop that kept top taxation high without any further shock and without a change in underlying preferences. Section 5 formalizes this through the framework of Bénabou (2000).

³Throughout, “war-active” countries are those that sustained military mobilization of at least two percent of their population during the war; “non-war-active” countries are those that did not, whether they stayed out of the conflict, declared war without mobilizing on that scale, or were invaded and occupied without mobilizing. The non-war-active group is therefore not a set of pure non-participants but the full set of countries against which the war-active countries are compared. Section 3 sets out the classification.

We contribute to three literatures. To the literature on warfare, taxation, and fiscal capacity, we show that the war raised top rates immediately, held them high for decades, and that the effect then faded. To the literature on long-run institutional persistence, we show that a single event reshaped a core fiscal institution, the income tax, for decades. To the literature on cross-country differences in tax progressivity, we offer a historical explanation for differences usually taken as given: much of the cross-country variation in progressivity observed over the second half of the twentieth century was produced by this one shock.

Our first contribution is to the literature on warfare, taxation, and fiscal capacity, whose closest precedent for our question is [Scheve and Stasavage \(2010\)](#), who establish that mass mobilization raised the demand for progressive taxation but neither isolate a single war nor trace its effect over time — the two things this paper adds. World War II is an especially suitable case: the conflict sits at the peak of the era in which mass mobilization could raise top taxes ([Onorato et al., 2014](#)), and its belligerents financed it substantially from domestic taxation rather than foreign borrowing, the mode of war finance that builds fiscal capacity ([Queralt, 2019](#)). We estimate that war entry raised the top rate by about 38 percentage points within five years and then, to our knowledge for the first time, trace the full path of that effect across the following decades—the long-run dimension on which the rest of the paper builds.

Our second contribution is to the literature on long-run institutional persistence. That literature identifies the conditions under which a transitory shock breaks institutional inertia ([Acemoglu and Robinson, 2008](#))⁴ and documents cases where one has ([Acemoglu et al., 2011](#)), but the shocks are rarely sharp enough, or the outcome cleanly enough measured, to trace the full trajectory. World War II—among the twentieth century’s largest compressions of inequality ([Scheidel, 2017](#))—and the top rate provide that clean case: a single event reshaped a core fiscal institution for decades. Applying the same design to World War I yields a similar but smaller effect, a suggestive sign that the pattern is not specific to one war. We also offer a mechanism distinct from the preference-based account of [Scheve and Stasavage \(2010\)](#), in which the war raised the demand for fairness directly: in ours, preferences are unchanged, and a large shock to the wealth distribution moves the economy to a more progressive equilibrium that then sustains itself, a mechanism we formalize in [Section 5](#).

Our third contribution is to the literature on cross-country differences in tax progressivity.

⁴[Acemoglu and Robinson \(2008\)](#) characterize these as shocks that alter the distribution of political power, the identity of the contending groups, and the stakes over which they bargain—all three of which mass mobilization changed at once.

Much of the theory here is normative: a large literature on optimal taxation asks how progressive taxes ought to be set (Diamond and Mirrlees, 1971; Judd, 1985; Chamley, 1986; Chari and Kehoe, 1999), but says little about why observed progressivity differs so sharply across countries. Where progressivity is explained at all, it is usually through present-day forces—the preferences of the median voter (Meltzer and Richard, 1981), optimal-tax considerations (Piketty et al., 2014), and competition for mobile capital (Keen and Konrad, 2013)—or modelled theoretically with limited empirical evidence (Persson and Tabellini, 2000; Yared, 2010). These accounts ask what sustains progressivity in the present rather than where today’s cross-country differences came from. We offer a historical answer: much of that variation, observed over the second half of the twentieth century, can be traced to World War II rather than to present-day forces alone.

The closest precedent for this paper is Scheve and Stasavage (2010), which identifies the effect of mass mobilization on top rates around the First World War but never isolates the Second. The Second World War enters their analysis only inside a pooled indicator that combines it with the First World War and other conflicts, so their work yields no estimate of what the Second World War alone did to progressive taxation. We identify the Second World War on its own, estimating the effect year by year after each country’s entry, and on a panel of thirty-five countries — far larger than the difference-in-differences across eight countries in Scheve and Stasavage (2010), or the twenty-country database of Scheve and Stasavage (2016), that this literature has relied on. Tracing the full trajectory over the seven decades that follow is the central result of the paper. That trajectory shows what their design cannot: the Second World War produced not just an immediate rise in top rates but a shift that remained statistically distinct from zero for at least five decades, with point estimates staying positive while declining thereafter as top rates fell across countries after 1980. That later decline coincides with the broad post-1980 fall in top rates, which Saez and Zucman (2019) describe as a deliberate policy turn rather than a forced response to economic integration; Frizell (2025), analyzing conflict-affected countries from 1960 to 2020, likewise finds that the link between war and progressive taxation itself disappeared after the end of the Cold War. The war’s effect on progressive taxation was therefore long-lived rather than a wartime episode, a claim only the isolated, full-path estimate can support—and providing that estimate is one of this paper’s central contributions.

The paper proceeds as follows. Section 2 sets out the historical background, how World War II’s fiscal and political demands bore on top taxation. Section 3 describes the data. Section 4 presents

the empirical strategy, the main results, and the robustness checks. Section 5 develops the equilibrium framework that interprets the persistence. Section 6 concludes.

2 Historical Background

2.1 The fiscal shock and its financing

The Second World War drove public spending in the belligerent states to levels without peacetime precedent. Total war placed every belligerent under fiscal demands — for armies, equipment, and the industrial effort behind them — on a scale their pre-war tax systems had never been designed to raise. Existing revenue tools, built for peacetime budgets, could not close the gap. Governments were therefore pushed toward extraordinary measures: not marginal adjustments to existing taxes, but a structural expansion of the state’s capacity to extract revenue.

Where that revenue was found shaped the fiscal institutions the war left behind. Financing a war by taxing the domestic population, rather than by borrowing abroad, is what builds a state’s capacity to assess and collect revenue ([Queralt, 2019](#)). The belligerents of the Second World War met the bulk of the war’s cost from domestic resources ([Harrison, ed, 1998](#)), and within that effort the progressive income tax was the lever capable of raising large sums quickly from those most able to pay, alongside domestic borrowing.

2.2 Mass mobilization and the demand for progressive taxation

Mass mobilization generated political pressure to tax the rich, distinct from the war’s fiscal need for revenue. By conscripting ordinary citizens into the war effort on a mass scale, total war produced a widely shared demand that owners of wealth bear a comparable burden at home ([Scheve and Stasavage, 2010](#)), and heavier progressive taxation was the form that demand took. What drove this demand was the breadth of mobilization—the share of the population drawn into military and war service—rather than the intensity of combat, so it arose even in countries that saw little fighting on their own soil.

Two features of how the war arrived shaped its fiscal impact. Mass mobilization was a different fiscal event from the rearmament that preceded it, and the timing and scale of that mobilization were largely settled by events outside each country’s control.

Rearmament through the late 1930s was financed largely by borrowing rather than by broad

increases in the taxation of top incomes (Tooze, 2006). The large, structural increases in top rates came with the mass mobilization of 1939–1941 (Scheve and Stasavage, 2016), whose fiscal demands far exceeded those of rearmament. The rise in top rates thus moved with the scale of wartime mobilization rather than tracking the earlier, debt-financed rearmament.

Whether and when a country was drawn into the war was settled largely by events outside its own control. British policy through the late 1930s sought to preserve peace, and the guarantee to Poland of March 1939 was meant to deter a war rather than to wage one.⁵ Deterrence failed: the invasion of Poland drew Britain and France in within days, and the attack on Pearl Harbor in December 1941 forced the United States to enter (Weinberg, 1994). Even the war’s initiators misjudged its course, since in 1938 neither Germany nor its opponents foresaw the scale the conflict would reach (Tooze, 2006). A country’s entry into the war, and the fiscal shock that came with it, was therefore settled from outside and could not have been anticipated.

2.3 The postwar settlements and their persistence

The wartime increase in progressive taxation outlasted the war because the postwar welfare state was built on it. Total war reshaped social policy so durably that the wartime expansion of public provision created a postwar coalition unwilling to let the new fiscal commitments lapse (Titmuss, 1950). Across Western democracies the legislative consolidation of the postwar welfare state clustered in the years immediately after the war (Obinger and Schmitt, 2020); the elevated, progressive taxation built up during the war financed that settlement and was entrenched along with it rather than rolled back at the war’s end. That the settlement embedded these increases, however, does not by itself explain why they endured for decades rather than eroding once the war receded.

The settlement held because the war had moved the underlying political economy, not merely the tax schedule. A transitory shock leaves lasting institutional change only when it alters the distribution of power, the identity of the contending groups, and the stakes they contest (Acemoglu and Robinson, 2008). Mass mobilization did all three at once: it raised the political weight of ordinary citizens, eroded the prewar concentration of wealth in several belligerents, and narrowed the resources available to those who would resist redistribution.

⁵The clearest expression of this approach was the personal diplomacy of September 1938, when Chamberlain travelled to Germany three times to negotiate directly with Hitler over Czechoslovakia, concluding the Munich Agreement and returning to London to declare “peace for our time.” That settlement collapsed when Germany seized the remainder of Czechoslovakia in March 1939, and the guarantee to Poland that followed committed Britain to the war it had spent the previous year trying to avert (Weinberg, 1994).

Breaking the old equilibrium did not, on its own, install a durable replacement. The elevated rates moved remarkably little once the war ended: across the major belligerents they were sustained for three decades through repeated changes of government and shifts in the ideological climate, not only in the immediate postwar years (Scheve and Stasavage, 2016). Bénabou (2000) supplies the structural reason such durability is possible: inequality and redistribution can support more than one stable configuration, each sustained by the distribution of income and political influence that prevails within it. Once the war had compressed wealth and broadened the political weight of ordinary citizens, the more redistributive of these configurations became self-sustaining, reproducing without the continuing pressure of mobilization. The persistence is therefore a property of the equilibrium the war selected, not of any lasting change in preferences—the mechanism Section 5 formalizes.

This equilibrium concerns the progressivity of taxation, which is a separate matter from the size of the fiscal state. That the fiscal state stayed large has a familiar explanation: war pushes public expenditure to a permanently higher level — reconstruction, veterans’ pensions, war-debt service — and revenue rises to match (Peacock and Wiseman, 1961; Higgs, 1987). But a higher revenue requirement can be met by taxing broadly rather than progressively, so this ratchet accounts for the size of the postwar fiscal state, not the progressivity of the taxes that financed it. It is therefore a complement to the account developed here, not a rival: the ratchet fixes how much revenue the state needed, while this paper explains why so much of that revenue kept coming from progressive taxation.

2.4 Precedent: the First World War

The persistence this paper identifies extends beyond the Second World War. If mass mobilization shifts the political economy toward sustained progressive taxation, the same should be visible after the First World War, the other total-mobilization conflict of the century. It is: the earlier war produced a comparable rise in progressive taxation that likewise outlasted the fighting. The First World War therefore serves as a second setting in which the mechanism this paper traces can be observed.

The First World War was a similar shock for the same reason the Second was: a war of mass mobilization, financed increasingly through the income tax. Top marginal rates were pushed to levels without precedent, and much of the population was drawn into the tax for the first time (Torregrosa-

Hetland and Sabaté, 2022). The wartime increases were partly rolled back afterward, but rates and revenues settled well above their pre-war levels rather than returning to them (Torregrosa-Hetland and Sabaté, 2022; Peacock and Wiseman, 1961). The same shock that raised progressive taxation during the war thus left a fiscal system permanently changed once it ended.

This paper examines the First World War directly. Applying the same event-study design to the earlier conflict yields a smaller but direction-consistent increase in top rates, reported in Section 4. The exercise is necessarily more limited than the main analysis: fewer countries have usable tax data for the period, and the years after 1918 are soon crowded by the Depression and the Second World War, leaving too short a window to trace persistence over the long horizons studied here.

2.5 The post-1980 erosion of top rates

The elevated top rates the war produced were eventually pushed back by forces that had nothing to do with the war. From the early 1980s a new policy consensus took hold across the advanced economies, holding that high top rates discouraged effort and investment, and conservative governments led the turn. In the United States the Reagan administration cut the top federal rate from 70 percent in 1980 to 28 percent by 1988; in the United Kingdom the Thatcher government cut the top rate from 83 to 40 percent over the same decade; and other countries followed (Saez and Zucman, 2019). Tax competition between countries reinforced the downward pressure (Genschel and Schwarz, 2011). This was a deliberate change in policy, not a forced response to economic integration (Saez and Zucman, 2019).

Because these cuts pulled top rates down across war-active and non-war-active countries alike, the post-1980 decline is a later, common force, distinct from the war’s own effect; the empirical analysis in Section 4 holds the two apart.

3 Data

3.1 The top marginal rate and the 35-country panel

The outcome studied in this paper is the top marginal rate of the personal income tax. The data come from the Comparative Income Taxation Database (Genovese et al., 2016), which records, for each country and year, the highest statutory marginal rate levied by the national government on personal income, and the law that sets it. Where surcharges or special levies form part of the income

tax, the recorded rate combines them; net wealth taxes and sub-national taxes are excluded.

The Comparative Income Taxation Database covers twenty countries, the sample that [Scheve and Stasavage \(2016\)](#) and subsequent work have drawn on. This paper extends it into a new panel of thirty-five countries, adding fifteen by drawing top-rate series from national tax legislation, statistical offices, and the historical public-finance literature. The additional countries widen the range of wartime experience the data contain, and in particular enlarge the set of countries that were not militarily active in the war. That larger and more varied set of control countries is what allows the war’s effect to be estimated more precisely than the original twenty would permit.

The fifteen additional countries draw on sources of two broad kinds. For some, the top rate comes from national tax legislation and official statistical agencies, such as Mexico’s INEGI. For others, it comes from the historical public-finance and top-incomes literature, including the country studies of the World Inequality Database and national fiscal histories. Each series was assembled from the most authoritative source available for that country, and [Appendix 8](#) lists the source used for every country.

Top rates vary widely across the panel, both between countries at any moment and within countries over time. The cross-country mean of the top rate rises through the war and early postwar decades — from 28 percentage points before the war to 44 during it and 53 in 1946–1970 — and then recedes to 48 in the late twentieth century and 36 in the 2000s ([Table 1](#)). Within-era dispersion stays wide throughout: the range between the lowest and highest national top rates exceeds 70 percentage points in every era from 1929 through 2000 and remains above 40 percentage points in the 2000s. The standard deviation across countries within each era is correspondingly large, on the order of one-third to two-thirds of the period mean, so the upward shift in averages does not amount to a convergence in rates. This combination of a shared broad trajectory and wide within-era dispersion is the variation the analysis exploits.

3.2 Classifying countries by wartime mobilization

The design compares countries that mobilized for the war with those that did not. Following [Scheve and Stasavage \(2010\)](#), whose mechanism ties the demand for progressive taxation to mass mobilization, we code a country as *war-active* if it sustained military mobilization of at least two percent of

Table 1: Top marginal rate across the panel, by era

Era	N	Countries	Mean	SD	Min	Max
Pre-war (1929–1938)	333	35	27.61	18.35	4.80	79.00
War (1939–1945)	240	35	44.20	27.10	4.80	97.75
Early postwar (1946–1970)	858	35	52.61	22.30	8.00	97.50
Late twentieth century (1971–2000)	1050	35	47.77	16.34	9.00	97.75
Recent (2001–2010)	350	35	35.59	9.31	11.50	55.00
Full panel (1929–2010)	2831	35	45.06	20.71	4.80	97.75

Notes: N is the number of country-year observations with a non-missing top marginal rate. Countries is the number of distinct countries contributing to the era. Mean, SD, Min, and Max are computed across country-year observations and reported in percentage points.

its population during the war,⁶ and as *non-war-active* otherwise. The criterion is mobilization, not a declaration of war: several countries declared war without mobilizing on this scale, and they are among the controls. In one respect we apply the criterion more consistently than Scheve and Stasavage, who also require interstate-war participation and so treat high-mobilization neutrals such as Sweden and Ireland as non-participants; we code these two as war-active. Thirteen of the thirty-five countries are war-active and twenty-two are controls (Appendix 8 lists each).

Two further countries, Austria and Switzerland, cross the mobilization threshold but are kept as controls on data grounds. Austria’s top rate in 1938–1945 is German tax law applied to occupied territory rather than an Austrian fiscal decision; Switzerland’s wartime fiscal effort ran through a new federal levy rather than its top rate, so the recorded series does not capture it. Section 4.4.5 re-classifies these borderline cases and the result holds. A finer breakdown by wartime exposure — separating active combatants from countries occupied without mobilizing and from lower-mobilization or neutral participants — is reported in Appendix B.3.

3.3 Additional tax progressivity measures

A further feature of the tax code captures progressivity directly and complements the top income rate: the top statutory rate on large inheritances and estates. Like a high rate at the top of the income schedule, a high inheritance-rate ceiling places the heaviest burden on holders of large fortunes, so it captures how heavily the tax system bears on the transfer of accumulated wealth at the top of the distribution. The series is taken from Scheve and Stasavage (2012), whose data cover

⁶Mobilization is measured at its sustained wartime level, not its peak: a country that mobilized briefly and then demobilized, as Belgium did after its 1940 capitulation, is classified by the level it sustained through the war.

the top inheritance rate for nineteen of the sample countries; Table 2 summarizes it across eras. The inheritance rate is accordingly used as a complementary indicator rather than as a primary outcome, and its relationship to the top income rate is examined in Section 4.

Table 2: Top inheritance rate across the panel, by era

Era	N	Countries	Mean	SD	Min	Max
Pre-war (1929–1938)	184	19	18.45	15.81	0.00	76.00
War (1939–1945)	133	19	28.59	20.81	0.00	77.00
Early postwar (1946–1970)	475	19	37.77	24.15	0.00	90.00
Late twentieth century (1971–2000)	570	19	32.77	22.68	0.00	80.00
Full panel (1929–2000)	1362	19	32.17	23.06	0.00	90.00

Notes: Top statutory marginal rate on inheritances and estates, from [Scheve and Stasavage \(2012\)](#), for the 19 sample countries with usable estate-tax records. N is the number of country-year observations with a non-missing rate; the series ends in 2000, so no observations fall in 2001–2010. Mean, SD, Min, and Max are in percentage points.

3.4 The redistribution wedge and pre-tax inequality

Beyond the tax schedule itself, the measure that bears most directly on progressivity is the redistribution wedge: the gap between pre-tax and post-tax inequality. It captures how much the tax-and-transfer system compresses the distribution, so a widening wedge is direct evidence that the system became more progressive.

The components come from the World Inequality Database: the top-10% and top-1% shares of national income and the Gini coefficient, each in a pre-tax and a post-tax version, with the wedge defined as the pre-tax value minus the post-tax value. Table 3 summarizes the pre-tax and post-tax series across eras. The pre-tax series cover the panel over the postwar period; comparable post-tax series begin only around 1980 for most countries, so the wedge is observable for too short a window to trace the war directly, and we use it as a supplementary, contemporary indicator rather than as a primary outcome. The relationship of these measures to the top rate is taken up in Section 4.

4 Empirical Analysis

We treat each war-active country’s entry as an event and ask, year by year after entry, how its top rate moved relative to the path implied by the non-war-active countries and by the country’s own pre-event dynamics. The trajectory the design recovers — a sharp rise at entry, a persistently

Table 3: Pre-tax and post-tax inequality measures across the panel

Measure	N	Countries	Coverage	Mean	SD	Min	Max
Top-10% share, pre-tax	2033	35	1929–2010	38.9	10.4	18.9	67.5
Top-10% share, post-tax	1158	35	~1980–2010	35.7	13.4	17.5	66.0
Top-1% share, pre-tax	2033	35	1929–2010	13.3	5.9	2.6	35.8
Top-1% share, post-tax	1158	35	~1980–2010	11.4	6.3	2.3	32.8
Gini, pre-tax	2033	35	1929–2010	49.8	10.6	28.6	76.5
Gini, post-tax	1136	35	~1980–2010	43.9	15.7	17.3	74.4

Notes: World Inequality Database income shares and Gini coefficients, pre-tax (`ptinc`) and post-tax (disposable income), for the 35 sample countries; shares and Gini are on a 0–100 scale. N is the number of non-missing country-year observations; post-tax series are available for most countries only from around 1980. The redistribution wedge analyzed in Section 4 is the pre-tax value minus the post-tax value.

elevated path for several decades, and a fade at the longest horizons — is what this section identifies as the war’s causal effect.

4.1 Cross-country evidence

The raw data already show the pattern the formal design will recover. In war-active countries the average top rate rises sharply around the time of war entry and stays elevated for decades, while in non-war-active countries it drifts at a much lower level until the post-1980 retreat of top taxation (Figure 2, panel b). The same shape is visible for the First World War on a smaller sample (panel a), where the war-active countries’ top rate lifts sharply during the war and stays higher through the interwar period. A fairly constant gap separates the two groups before each war — on the order of fifteen to twenty percentage points before the Second World War and a few percentage points before the First — and it widens sharply at war entry. The formal design takes up how this preexisting gap is held apart from the war’s effect; the averages here only set out the pattern the design then identifies.

4.2 Empirical strategy

We take a country’s war entry as the event and estimate its dynamic effect on the top rate using an event study with a lagged dependent variable. The specification is

$$\text{toprate}_{i,t} = \sum_{\tau \in \mathcal{T}} \beta_{\tau} \cdot \mathbf{1}[e_{i,t} = \tau] + \rho \cdot \text{toprate}_{i,t-1} + \delta \cdot \text{Post}_{i,t} \cdot \text{toprate}_{i,t-1} + \alpha_i + \varepsilon_{i,t}, \quad (1)$$

where $\text{toprate}_{i,t}$ is the top marginal income tax rate in country i in year t ; $e_{i,t}$ is event time, the year minus country i ’s war-entry year; $\mathbf{1}[e_{i,t} = \tau]$ indicates being τ years after entry; β_{τ} is the

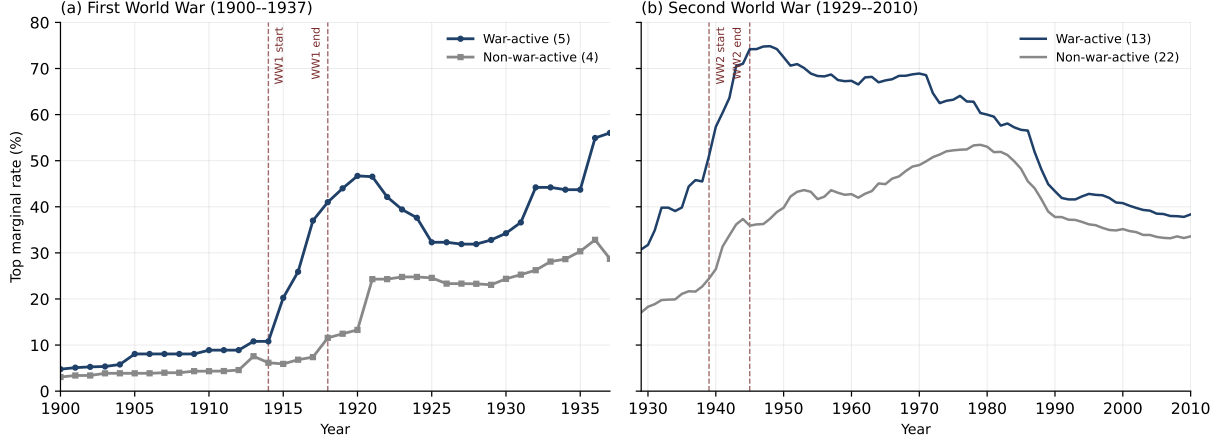


Figure 2: Average top marginal rate by group and year. Panel (a): First World War, on the nine-country sample of war-active countries (ITA, JPN, NZL, GBR, USA) and European-neutral controls (DNK, NLD, NOR, SWE). Panel (b): Second World War, on the full 35-country panel (13 war-active, 22 non-war-active). Each line is the unweighted cross-country mean of the top statutory rate within a group; vertical lines mark the start and end of each war.

per-period effect of war entry on the top rate τ years out; ρ is the persistence of the top rate, common to all countries; $\text{Post}_{i,t}$ is the post-entry treatment indicator, equal to one for a war-active country in the years on or after its war entry and zero otherwise, so it switches from zero to one within a country at entry; δ is the coefficient on the interaction of that indicator with the lagged top rate, $\text{Post}_{i,t} \cdot \text{toprate}_{i,t-1}$, and measures the change in persistence for war-active countries from entry onward; α_i are country fixed effects; and $\varepsilon_{i,t}$ is the residual. The event-time window is $\mathcal{T} = \{-5, \dots, -2\} \cup \{0, \dots, 71\}$, with $\tau = -1$ omitted as the baseline. Standard errors come from a wild cluster bootstrap on country, with 999 bootstrap replications drawing Rademacher weights on country clusters (Cameron et al., 2008).⁷

Top rates are institutionally sticky: tax systems adjust slowly, and a country’s rate next year is well predicted by its rate this year. This autocorrelation is common to all countries in the panel, and ρ measures it; the term δ then allows that persistence to differ for war-active countries after entry. Conditioning on the lagged rate does more than absorb this autocorrelation. Last year’s rate already reflects the slow-moving forces that produced it — a country’s fiscal capacity, its political institutions, the prevailing climate of opinion on taxation — so controlling for it nets out the persistent part of those forces, and the post-entry movement is measured relative to where

⁷The wild cluster bootstrap is the recommended small-sample correction when the number of clusters is small (Cameron et al., 2008). Setting the number of replications to $B = 999$ follows standard practice: a bootstrap p -value is computed as $(1+r)/(B+1)$, where r counts the replications whose statistic is more extreme than the one observed in the data, so $B+1 = 1,000$ places conventional significance thresholds exactly on a replication.

each country’s rate had recently been rather than to its fixed level alone.⁸ The first-order lag is the simplest structure that captures the persistence, and lag-length sensitivity in Section 4.4 shows the persistence estimate and the cumulative trajectory are stable across AR(1)–AR(4). The design follows the precedent of Acemoglu et al. (2019), who use the same lagged-dependent-variable structure to estimate the long-run effect of democracy on GDP per capita, another highly persistent outcome.

The interaction $\text{Post}_{i,t} \cdot \text{toprate}_{i,t-1}$ lets the war change the dynamics of the tax system, not only its level. For a war-active country, persistence is governed by ρ before entry and by $\rho + \delta$ from entry onward, so δ measures the war-induced change in the persistence of the tax institution, not an increment to a fixed level. The distinction matters for the paper’s argument: the war did not merely raise top rates, it altered the speed at which the tax system adjusts, an institutional change that is part of why the higher rates endured. Without the δ term the specification would force the war to move only the level and leave the adjustment dynamics unchanged, ruling out the institutional shift that the equilibrium mechanism of Section 5 describes.

With these components in place, the cumulative effect at horizon $\tau \geq 0$ is the recursive object

$$\Delta_\tau = \beta_\tau + (\rho + \delta) \cdot \Delta_{\tau-1}, \quad \Delta_{-1} \equiv 0, \quad (2)$$

where Δ_τ is the total change in the top rate τ years after entry that the model implies: each year’s new per-period impulse β_τ is added to the carried-forward effect of all earlier years, $(\rho + \delta) \Delta_{\tau-1}$. The high estimated persistence is what turns the initial increase into a path that stays elevated for decades. The boundary condition $\Delta_{-1} \equiv 0$ anchors the path at the entry-baseline year $\tau = -1$, so Δ_τ measures the deviation of a war-active country’s top rate from the path it would have followed without the war, the counterfactual traced by the non-war-active countries.

The pre-event window is set by the economics of the period. It starts six years before each treated country’s entry, roughly the midpoint between the Great Depression (1929–1933, itself a major structural break in tax data) and the 1939 entry of the European belligerents. A longer pre-window would reach back into the Depression and contaminate the pre-period with its tax changes; a shorter one would leave too little pre-event variation to anchor on. Alternative window lengths in

⁸A lagged dependent variable absorbs unobserved determinants of the top rate only insofar as they are carried in the previous year’s rate; it does not control for arbitrary contemporaneous shocks. The alternative specifications in Section 4.4 address those directly, adding year fixed effects and time trends.

Section 4.4 do not materially affect the main findings.

The design identifies the war’s effect on the top rate under three assumptions — conditional parallel trends, no anticipation, and no bundled confounder — which we set out and assess below; the empirical evidence for each is collected in Section 4.4.

The first assumption is conditional parallel trends: absent the war, top rates in the war-active and non-war-active groups would have moved together, once each country’s fixed level and its persistence are accounted for. The country fixed effects absorb fixed cross-country differences in the level of top rates, so a country that taxes top incomes heavily throughout the panel does not by itself drive the estimate. The lagged dependent variable absorbs the persistent component of each country’s path, so a country already drifting upward before entry is not credited with a war effect on that account. Broad forces shaped top rates over the whole period — the spread of capital mobility, the secular growth of fiscal capacity — but these bore on the war-active and control countries alike, so they do not by themselves produce a divergence between them. The assumption has a direct test: whether the two groups were on parallel paths before entry. The pre-event coefficients reported below are jointly indistinguishable from zero, which is what conditional parallel trends implies and is evidence in its favor.⁹

A second assumption is no anticipation: top rates did not adjust ahead of war entry. The historical record supports this on two counts. First, rearmament in the late 1930s was financed largely by borrowing rather than by raising top rates, so the broad-based increases came only with mobilization. Second, the timing and scale of that mobilization were settled by events outside each country’s control — the Munich-to-Poland sequence that drew Britain and France in, the Pearl Harbor attack that forced the United States in — and even the war’s initiators misjudged its eventual scale. A pre-entry adjustment would have required tax authorities to foresee both the war and its fiscal magnitude, which the events of 1938–1941 gave no government reason to expect.¹⁰

The third assumption concerns omitted variables. An omitted determinant of top rates biases the estimate only if it is correlated with the timing of war entry; a variable that moves top rates but is unrelated to war participation enlarges the residual variance without biasing the estimated

⁹What distinguishes the war-active group is mass mobilization, the source of variation that the historical account in Section 2 describes.

¹⁰Other studies likewise treat the Second World War as an exogenous shock for identifying postwar outcomes — labor-market and human-capital effects (Ichino and Winter-Ebmer, 2004; Akbulut-Yuksel, 2014), innovation (Gross and Sampat, 2023), and the broader compression of inequality (Scheidel, 2017) — relying on the same exogeneity of war timing.

effect. The variables that do coincide with entry — postwar reconstruction, Cold-War alliance and military commitments, and the postwar growth boom — are themselves largely consequences of war participation rather than independent causes, and to the extent they follow from entry the event-time dummies already absorb part of their effect.¹¹ The Cold-War concern in particular is addressed directly in Section 4.4 by restricting the sample to data through 1970, decades before late-Cold-War military burdens peaked, with the effect fully present at that cutoff.

These assumptions also receive statistical support in Section 4.4: a joint pre-trend test for parallel trends, an extended pre-window for anticipation, and sample-period truncation for Cold-War-era confounds. A heterogeneity-robust estimator for staggered entry and a sensitivity analysis for parallel-trends violations point the same way and are reported in the appendix.

4.3 Results

The cumulative effect rises sharply at entry, stays large for several decades, and becomes less statistically significant at the longest horizons (Figure 3; Table 4). Five years after entry the war-active country's top rate is about 38 percentage points above the path the model implies without the war ($\Delta_5 = +37.6$); a decade out it is about 40 ($\Delta_{10} = +40.2$); three decades out, about 42 ($\Delta_{30} = +42.3$). The cumulative effect remains statistically distinct from zero through roughly five decades ($\Delta_{50} = +19.6$, still significant) and then loses significance as its confidence interval widens ($\Delta_{70} = +12.3$, no longer distinct from zero). The point estimate stays positive throughout, but the estimates become less statistically significant the further the horizon is from entry. Because the cumulative effect is built up recursively, its confidence interval at a given horizon compounds the sampling uncertainty in each per-period coefficient and in the estimated persistence; that uncertainty accumulates as the horizon lengthens, and with only thirteen war-active countries the long-horizon intervals are correspondingly wide. At the longest horizons the war-active sample also thins, since only the earliest entrants are observed that many years after entry.

¹¹To the extent any such factor operated independently of war participation, the country fixed effects and the lagged dependent variable absorb each country's persistent level and dynamics, and the alternative specifications in Section 4.4 add common trends and year fixed effects that sweep out variables shared across countries; the point estimates shift, but the main picture — a sharp rise followed by a persistent, multi-decade effect — holds. The postwar growth boom affected both groups and so does not generate a divergence between them. Spending tied to reconstruction or alliance commitments raises revenue needs broadly rather than progressively, so it does not raise the top rate in particular and cannot account for the estimated effect.

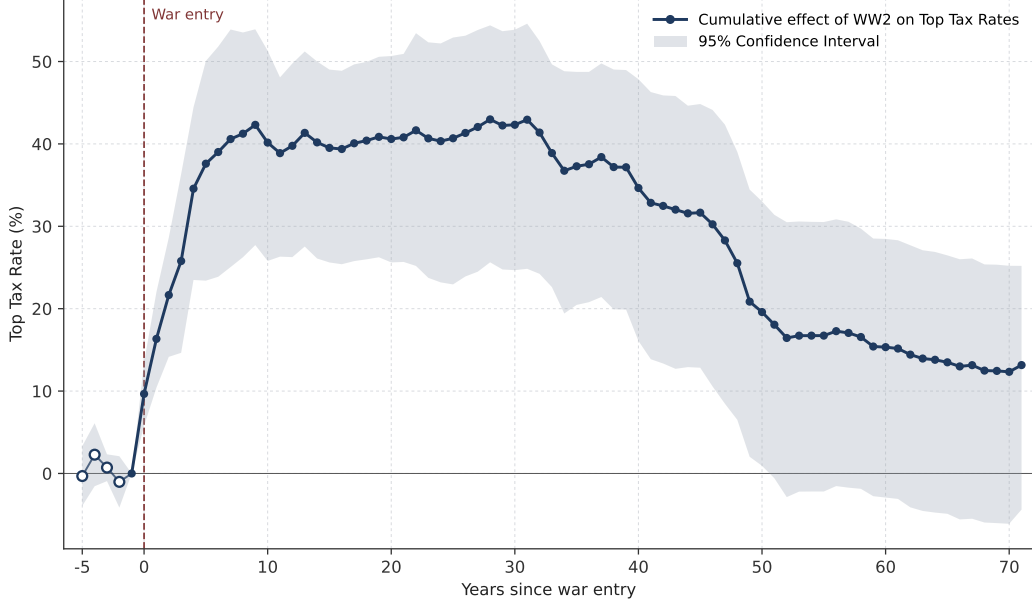


Figure 3: Cumulative effect of WW2 entry on the top marginal rate. The solid line traces Δ_τ over event time; the shaded band is the wild cluster bootstrap 95% confidence interval (999 bootstrap replications). The vertical line at $\tau = 0$ marks war entry; hollow markers at $\tau \in \{-5, \dots, -2\}$ are pre-event placebos. Source: the main specification, 35-country sample, $N = 2,767$.

The pre-event coefficients are economically small and jointly indistinguishable from zero, evidence in favor of the two groups having been on parallel paths before entry.¹² Each individual pre-event per-period coefficient lies within 2.3 percentage points of zero, with a confidence interval that straddles it. In the entry year the per-period coefficient jumps to $\beta_0 = +9.66$ percentage points, already three times the largest pre-event coefficient in absolute value — the immediate fiscal demand of the war showing up in top rates the year mobilization begins.

The persistence parameters underpinning this trajectory are worth reading on their own. The common persistence of the top rate, estimated across the whole panel, is very high, $\hat{\rho} = 0.95$: this year's rate is almost fully carried into next year's, so top rates change slowly. For war-active countries, entry lowers that persistence slightly, $\hat{\delta} = -0.06$, leaving post-entry persistence at $\hat{\rho} + \hat{\delta} = 0.89$ — still high, but below the common value. That the war reduced persistence rather than raising it may seem counterintuitive, but it follows from how the war moved tax policy: rates that

¹²A joint test of the four placebo leads $\tau \in \{-5, -4, -3, -2\}$ does not reject the hypothesis that they are jointly zero, returning a centered wild cluster bootstrap p -value of 0.945, far above conventional rejection thresholds.

ordinarily changed very little adjusted sharply during the war, as governments raised them to meet wartime fiscal demands and reset them in the postwar settlements. The war, in other words, reduced the stickiness, or inertia, of the tax system: for a war-active country the top rate became, over the war and the years that followed, less tightly anchored to its own past than the common persistence implies. The two parameters together describe an institutional change rather than a change in level alone: the war raised the top rate through the entry-year jump and also altered the dynamics by which the rate evolves, the shift the equilibrium account in Section 5 formalizes. Because post-entry persistence remains close to one, the entry-year jump is not eroded quickly but carried forward into a path that stays elevated for decades.

The cumulative trajectory grows over the first decade — $\Delta_5 = +37.6$, $\Delta_{10} = +40.2$ percentage points — because the immediate β_0 feeds forward through the high estimated persistence and the early post-event β_τ coefficients add to it (Table 4). The trajectory then plateaus at roughly $\Delta_\tau \approx +40$ percentage points through the first three decades and only begins to decay around $\tau = 40$ ($\Delta_{40} = +34.7$, $\Delta_{50} = +19.6$ percentage points). The decay sets in at the horizons that reach into the 1980s and beyond, when top rates fell broadly across countries (Section 2). This later decline does not work against the estimate as an uncontrolled confounder: it is common to both groups, so the comparison nets it out. The war set top rates on a higher path, while the post-1980 decline is a separate, common force — one that the year-fixed-effects specification in Section 4.4 absorbs directly, by removing movements common to all countries within a year, and still leaves a positive effect at long horizons.

Table 4: Cumulative effect Δ_τ at standard horizons

τ	Δ_τ (pp)	95% CI low	95% CI high
0	+9.66	+5.77	+13.49
5	+37.60	+23.40	+50.06
10	+40.16	+25.79	+51.31
15	+39.52	+25.61	+49.02
20	+40.59	+25.62	+50.67
30	+42.32	+24.67	+53.86
40	+34.65	+16.09	+47.86
50	+19.59	+0.93	+33.02
70	+12.34	-6.11	+25.21

Notes: Recursive AR(1) cumulative $\Delta_\tau = \beta_\tau + (\rho + \delta)\Delta_{\tau-1}$, $\Delta_{-1} \equiv 0$, from the main specification (equation 1): 35-country sample (13 treated, 22 controls), $K = 6$ pre-window, $N = 2,767$. Confidence intervals from the wild cluster bootstrap on country (999 bootstrap replications, Rademacher weights).

4.4 Robustness

This section subjects the trajectory to four kinds of stress test — alternative inference methods, alternative specifications, alternative estimators, and alternative outcomes and country classifications — and the central result survives all of them: the cumulative effect stays positive, large, and statistically distinct from zero through at least three decades, with point estimates close to the main estimate.

4.4.1 Alternative inference methods

A small panel raises the question of how to compute standard errors, because with few clusters the way they are computed — the assumed error structure and the asymptotic approximation — can matter more than the point estimate itself. We therefore report the cumulative effect under seven inference methods, ranging from the wild cluster bootstrap recommended for few clusters to estimators built for large panels (Figure 4; Table 5). The question is the horizon out to which the effect stays distinct from zero. The effect is significant at three decades under all seven methods, and at five decades under five of the seven: the wild cluster bootstrap, country-clustered, year-clustered, two-way-clustered, and OLS intervals all exclude zero at Δ_{50} , while only Driscoll–Kraay and the wild block bootstrap do not. At Δ_{70} no method reaches significance. The methods that lose significance earliest are those designed for large panels, so the loss of significance at $\tau = 50$ and $\tau = 70$ reflects the size of the sample — 35 countries, 13 treated — rather than a vanishing effect.

Table 5: Inference comparison: 95% confidence intervals under seven SE methods

Method	Δ_{30} 95% CI	Δ_{50} 95% CI	Δ_{70} 95% CI
OLS iid	[+25.89, +58.76] *	[+2.90, +36.28] *	[-4.39, +29.06]
Country-clustered	[+27.37, +57.27] *	[+3.28, +35.90] *	[-3.36, +28.04]
Year-clustered	[+23.19, +61.46] *	[+0.45, +38.74] *	[-6.60, +31.27]
Two-way clustered	[+24.17, +60.47] *	[+0.43, +38.76] *	[-6.59, +31.27]
Driscoll–Kraay	[+20.20, +64.45] *	[-2.72, +41.91]	[-9.79, +34.47]
Wild cluster bootstrap	[+24.86, +54.43] *	[+0.56, +33.65] *	[-6.33, +26.05]
Wild block bootstrap	[+17.11, +67.54] *	[-6.52, +45.70]	[-14.14, +38.82]

Notes: The point estimate is the same under every method — $\Delta_{30} = +42.32$, $\Delta_{50} = +19.59$, $\Delta_{70} = +12.34$ percentage points — so only the 95% confidence intervals are reported. Intervals excluding zero (significance at the 5% level) are marked *.

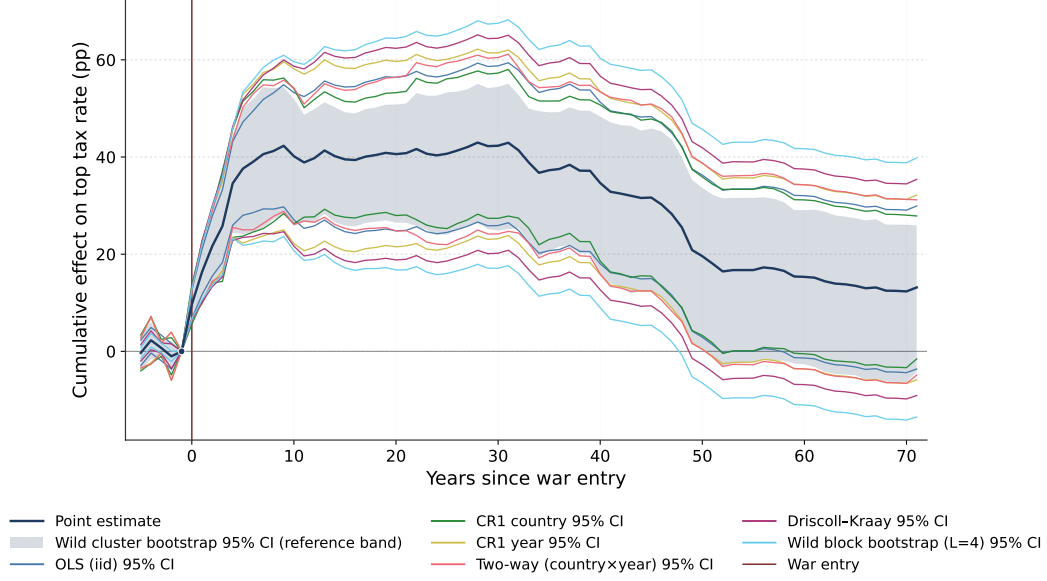


Figure 4: Cumulative effect Δ_τ with 95% confidence intervals under seven inference methods. The point estimate is the same for every method; the bands show how the interval width varies across them.

4.4.2 Alternative specifications

The main specification captures the slow evolution of top rates through a lagged dependent variable. We re-estimate the effect with alternatives that model that evolution differently — year fixed effects, common time trends, and longer lag structures — and each yields the same large positive effect at entry that persists for decades.

A first concern is that some common, time-varying force — the post-1980 retreat of top taxation, Cold-War fiscal pressures, or any other shock shared across countries — drives the post-entry path. Adding a full set of year fixed effects λ_t to equation (1) absorbs any such common annual shift into year dummies rather than letting it load on event time:

$$\text{toprate}_{i,t} = \sum_{\tau \in \mathcal{T}} \beta_\tau \cdot \mathbf{1}[e_{i,t} = \tau] + \rho \cdot \text{toprate}_{i,t-1} + \delta \cdot \text{Post}_{i,t} \cdot \text{toprate}_{i,t-1} + \alpha_i + \lambda_t + \varepsilon_{i,t}, \quad (3)$$

where λ_t are year fixed effects and all other terms are as in equation (1). Year fixed effects are the most demanding alternative here.¹³ The point estimate falls below the main estimate, but the

¹³War entry is concentrated in just three cohorts (1939, 1940, and 1941), so the event-time indicators are nearly collinear with the year dummies covering those years. With the sample small, the year dummies then absorb part of the very war effect being identified, damping the estimate rather than merely controlling for shocks common across countries.

effect stays positive across the trajectory and remains statistically distinct from zero at short and mid horizons.

Under this specification the cumulative effect keeps the sign and broad magnitude of the main estimate across the trajectory: $\Delta_5 = +26.4$, $\Delta_{15} = +26.3$, and $\Delta_{30} = +26.3$ percentage points $[-1.2, +46.1]$ (Figure 5). The confidence interval widens with the horizon as the year dummies absorb more of the variation, so the effect is statistically distinct from zero at short and mid horizons and its interval just reaches zero by three decades. The effect is attenuated relative to the main estimate but stays positive across the whole path.

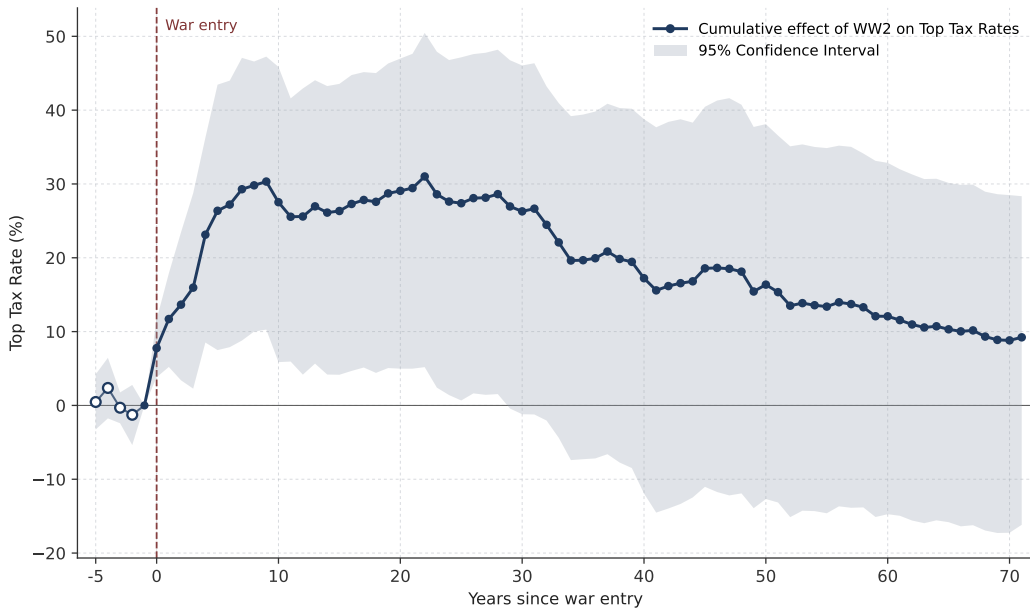


Figure 5: Cumulative effect Δ_τ under the main specification augmented with year fixed effects (equation 3). The shaded band is the 95% confidence interval.

A related concern is that the estimated persistence merely reflects war-active countries cutting their top rates more slowly after 1980, rather than a genuine wartime effect. Truncating the sample addresses this directly: restricting the panel to data through 1970, 1980, 1990, or 2000 yields Δ_{30} of +52.9, +52.6, +44.1, and +41.7 percentage points respectively, against the full-sample +42.3. The effect is largest at the 1970 cut — decades before any post-1980 turn in tax policy — which is the opposite of what the slow-cutting story predicts, and the pre-trend test passes at every cutoff.

A stricter alternative replaces the lagged dependent variable with a common parametric trend in calendar time — a quadratic or linear path shared across countries — that absorbs the slow

movement of top rates directly rather than through persistence. Because year fixed effects are the more flexible way to absorb shocks common across countries, these parametric trends are a special case of the same idea; we report them, with a figure comparing the specifications’ per-period paths, in Appendix B.4, where the effect again survives.

Longer lag structures reproduce the result. Re-estimating the model with two, three, or four lags of the top rate yields Δ_{30} of +44.1, +44.8, and +45.1 percentage points respectively, against the AR(1) main estimate +42.3; the estimated persistence is stable across lag orders, $\rho + \delta = 0.88\text{--}0.89$.

4.4.3 Alternative estimators

The result does not hinge on the pooled event-study estimator: three alternative estimators all reproduce the trajectory. The first is a leave-one-country-out re-estimation: dropping each of the 35 countries in turn leaves the whole trajectory close to the main estimate, with Δ_{30} ranging over $[+36.2, +45.2]$ percentage points around a median of +42.4, the most influential single country being Italy (Table 9). The second is a model-free check — a simple difference in group means at each horizon, with no dynamic structure imposed — which confirms that the early effect does not depend on the lagged-dependent-variable specification: the raw between-group gap is +16.4 percentage points at $\tau = 5$ and +13.3 at $\tau = 10$, both significant, then fades and turns negative at long horizons as top rates fall across both groups after 1980 — the common decline that the main specification holds apart by fixing the no-war counterfactual.

The third check computes the long-run effect a different way. The steady-state multiplier of Acemoglu et al. (2019) summarizes the entire post-entry response in a single number, $\beta_{LR} = \beta / (1 - \sum_j \gamma_j)$, estimated from the dynamic-panel specification $y_{i,t} = \beta \cdot \text{Post}_{i,t} + \sum_{j=1}^p \gamma_j y_{i,t-j} + \alpha_i + f(t) + \varepsilon_{i,t}$. The cost of this formulation is that it collapses the dynamics into one long-run figure and does not trace the effect year by year, as the main specification does; its value is that it reaches the same conclusion from a different direction. The within-OLS estimate gives β_{LR} in $[+25.6, +30.2]$ percentage points across lag orders $p = 1, \dots, 4$, with estimated persistence between 0.93 and 0.94 (Table 10). Even under this different formulation, then, the long-run effect is positive and economically large: it implies that in the long run a war-active country’s top rate settles roughly 25 to 30 percentage points above what a comparable non-participant’s would be.

Two further checks bear on the design’s identifying assumptions and are reported in full in the appendix; both confirm the result. The heterogeneity-robust estimator of Sun and Abraham

(2021) guards against bias from staggered entry and reproduces the per-period path almost exactly (Appendix B.1); the HonestDiD analysis of Rambachan and Roth (2023) shows that the effect survives violations of parallel trends as large as the pre-period suggests is plausible (Appendix B.2).

4.4.4 Progressivity beyond the top income rate

The war raised progressivity beyond the top income rate. Two further measures of how heavily the tax-and-transfer system bears on the top move the same way as the top income rate: the top statutory rate on inheritances, a second feature of the tax schedule, and the redistribution wedge, the gap between pre-tax and post-tax inequality (Table 6).

The top inheritance rate rose with the top income rate. Re-estimating the main specification with the top inheritance rate as the dependent variable, on the 19 countries for which it is available, gives a cumulative effect that rises over the first decade and stays elevated — $\Delta_5 = +13.4$, $\Delta_{15} = +24.2$, and $\Delta_{30} = +19.3$ percentage points, distinct from zero throughout under country-clustered standard errors.¹⁴ A second tax aimed at large fortunes thus moved with the top income rate, indicating the war steepened the schedule broadly rather than at a single margin.

The redistribution the tax-and-transfer system achieves also rose. Comparable post-tax series begin only around 1980, too short a window for the event study, so for the wedge we estimate a difference-in-differences between the war-active and non-war-active groups over the post-1980 period. The war-active effect is positive and distinct from zero for both the Gini-based wedge (+5.7 [+3.4, +8.1]) and the top-10% wedge (+2.6 [+0.9, +4.2]): the war-active countries compressed both the overall distribution and the top decile, through taxes and transfers, by more than they would have absent the war.

These additional measures point the same way: the war raised progressivity broadly, not only the single rate at the top of the income schedule. Each rests on a smaller sample than the main result — the inheritance series on 19 countries, the wedge on the shorter post-1980 window — so each carries less weight on its own, but together they are strong evidence that the war made the tax-and-transfer system more progressive overall.

¹⁴Excluding Italy, the most influential country, gives $\Delta_{30} = +17.0$ [+3.1, +31.0], still positive and distinct from zero.

Table 6: Progressivity beyond the top income rate

Outcome	Sample	Δ_5	Δ_{15}	Δ_{30}
<i>Panel A. Tax schedule; cumulative effect</i>				
Top marginal income rate (main estimate)	35 countries	+37.60	+39.52	+42.32
Top inheritance rate	19 countries	+13.38	+24.23	+19.27
<i>Panel B. Redistribution wedge (pre-tax – post-tax); difference-in-differences</i>				
Gini-based wedge	35 countries	+5.73	[+3.36, +8.09]	
Top-10% wedge	35 countries	+2.57	[+0.91, +4.22]	

Notes: Panel A: cumulative effect Δ_τ (percentage points) at five, fifteen, and thirty years after entry. Panel B: the war-active effect on the redistribution wedge (pre-tax minus post-tax inequality), estimated as a difference-in-differences over the post-1980 period (see text). All confidence intervals are country-clustered (CR1) at 95%.

4.4.5 Alternative country classifications

The classification rests on the two-percent mobilization threshold of Section 3.2. A few countries sit close to it, so if the result turned on exactly how they are coded it would be fragile; we therefore re-estimate the effect under deliberate reclassifications of the borderline cases (Table 7).

The borderline cases are the three countries discussed in Section 3.2: Sweden and Ireland, which the consistent mobilization rule codes war-active, and Austria, which is kept as a control. A strict variant returns Sweden and Ireland to the control group — the classification Scheve and Stasavage would use, since neither fought an interstate war — while an inclusive variant instead adds Austria to the war-active group, counting its Reich-administered wartime rate as a war-active observation; a third variant does both. Each variant re-estimates the full event study on the reclassified sample, so the cumulative trajectory is recovered exactly as in the main specification. All three leave the trajectory close to the main estimate and statistically distinct from zero at three decades — the strict variant, for instance, gives $\Delta_5 = +39.9$ and $\Delta_{30} = +45.4$, against the main +37.6 and +42.3 — so the result does not turn on how the borderline countries are coded. A finer breakdown by wartime-exposure category, which asks whether the effect extends beyond the active combatants, is reported in Appendix B.3.

Table 7: Borderline classification sensitivity: Δ_τ by variant

Variant (n_T)	Δ_0	Δ_5	Δ_{10}	Δ_{15}	Δ_{20}	Δ_{30}	Δ_{40}	Δ_{50}	Δ_{70}
Main specification (13)	+9.66	+37.60	+40.16	+39.52	+40.59	+42.32	+34.65	+19.59	+12.34
(a) Strict — drop SWE, IRL (11)	+9.20	+39.90	+42.77	+41.97	+43.51	+45.43	+38.86	+22.90	+17.24
(b) Inclusive — add AUT (14)	+8.98	+35.17	+37.85	+38.33	+38.61	+41.00	+34.77	+20.00	+13.30
(c) Combined — drop SWE/IRL, add AUT (12)	+8.55	+37.36	+40.55	+41.20	+41.81	+44.54	+39.57	+24.03	+18.90

Notes: Cumulative Δ_τ in percentage points; the sample is fixed at 35 countries in every variant, and only the war-active/comparison split changes. n_T is the number of war-active countries. Confidence intervals and pre-trend tests for each variant are reported in Appendix Table 11.

4.5 Summary and comparison with Scheve and Stasavage (2010)

The empirical analysis yields one main result with a set of checks around it. War entry raised the top marginal rate sharply — about 38 percentage points within five years — and the cumulative effect stayed near 42 points for three decades before fading after 1980, remaining statistically distinct from zero for roughly five decades. The trajectory is robust across inference methods, specifications, estimators, alternative progressivity measures, and country classifications, and the same design applied to the First World War yields a similar but smaller effect. We now set this result against the closest prior work.

The closest precedent is [Scheve and Stasavage \(2010\)](#), who first linked mass mobilization to progressive taxation: across the belligerents of the early twentieth century, countries that mobilized raised top rates by roughly 31 to 36 percentage points around the First World War, close to our own first-five-year estimate. We take that connection as our starting point and go substantially further — isolating a single war, tracing its effect across five decades, and doing so on a much larger and more varied panel — as the rest of this section sets out.

Their design, however, cannot isolate the Second World War, because the mobilization indicator pools several conflicts into a single coefficient. It is switched on for the two World Wars, the Franco-Prussian War, and the Korean War together, so the estimated effect answers whether mass mobilization in general raised top rates, not what the Second World War in particular did: a coefficient that the First World War, Korea, and the Franco-Prussian War also identify cannot deliver a Second-World-War-specific estimate. Their panel also ends in 1970, before the post-1980 decline, so it cannot trace how such an effect evolved over the following decades.

This paper supplies what their design leaves open: the effect of the Second World War alone on top rates, traced year by year across the more than five decades that follow. Estimating the effect separately for each year after a country’s entry — on the war-active/non-war-active classification of

Section 3.2, which applies the two-percent mobilization threshold consistently — isolates the Second World War from the other conflicts and recovers the full path: a rise of about 38 percentage points within five years, a plateau near 42 points for three decades, and a slow decline after 1980. Where the two studies overlap the magnitudes agree, and applying the same design to the First World War gives a smaller effect ($\Delta_5 = +30.4$, easing to $\Delta_{10} = +23.8$ percentage points), just below the range they report. The contribution is not merely to study one war rather than several. By isolating the Second World War and following it for half a century, we show that this single shock is largely responsible for the cross-country differences in tax progressivity that opened up in the decades after the war — differences that later work has often taken as given.

5 A Model of War and Progressive Taxation

This section gives the empirical findings an economic mechanism, building on the multiple-equilibria framework of Bénabou (2000) to show why a transitory shock can have lasting effects. The question it answers is why a one-time shock to inequality — the war — can produce changes in top tax rates that persist for decades without any change in preferences. The answer is multiple equilibria: a sufficiently large shock moves the economy between basins of attraction, and the equilibrium it selects then sustains itself under unchanged primitives.

5.1 The mechanism and its relationship to Bénabou (2000)

We build on the multiple-equilibria framework of Bénabou (2000), in which the redistribution a society chooses and the inequality it lives with are determined jointly, so that the same fundamentals can support more than one stable outcome. The logic is intuitive: a more equal distribution of wealth strengthens political support for redistribution, and more redistribution in turn keeps the distribution equal, so a large enough shock to inequality can move the economy from a low-redistribution outcome to a high-redistribution one that then sustains itself. We make two adaptations to Bénabou (2000). We simplify preferences to log utility, so the case for redistribution does not rest on a high coefficient of relative risk aversion; and we bring the framework to bear on an identified shock, the Second World War, that Bénabou never models. The next subsection sets out the model formally.

5.2 The model

The economy is populated by a continuum of dynasties indexed by $i \in [0, 1]$, with overlapping generations. Each adult lives one period, inheriting capital from the parent, supplying labour, producing output, paying taxes, and bequeathing capital to the child.

Preferences are log over consumption c_t^i , leisure, and the child's capital endowment:

$$U_t^i = (1 - \pi) \left[\ln c_t^i - \frac{(\ell_t^i)^\eta}{\eta} \right] + \pi \ln k_{t+1}^i, \quad (4)$$

where $\pi \in (0, 1)$ governs the strength of the bequest motive, $\eta > 1$ controls labour-disutility curvature (with Frisch elasticity $1/(\eta - 1)$), and ℓ_t^i is labour effort. Output is

$$y_t^i = z_t^i (k_t^i)^\alpha (\ell_t^i)^\delta, \quad (5)$$

with capital share α , labour share δ , and an idiosyncratic productivity shock z_t^i with $\ln z_t^i \sim \mathcal{N}(-v^2/2, v^2)$. The cross-sectional variance of log capital, $\sigma_t^2 \equiv \text{Var}(\ln k_t^i)$, is the state variable that summarises inequality.

Fiscal policy follows Bénabou's residual-progression scheme:

$$\hat{y}_t^i = (y_t^i)^{1-\tau_t} \cdot \tilde{y}_t^{\tau_t}, \quad (6)$$

where $\tau_t \in [0, 1]$ is the progressivity parameter and $\tilde{y}_t$ is set by budget balance, $\int \hat{y}_t^i di = \int y_t^i di$. The variance of log after-tax income is exactly $(1 - \tau_t)^2$ times the variance of log pre-tax income, which makes the redistribution channel transparent: $\tau_t = 0$ is laissez-faire and $\tau_t = 1$ is full equalisation.

Capital accumulates through two channels:

$$k_{t+1}^i = (k_t^i)^\theta \cdot (\rho \hat{y}_t^i)^{1-\theta}, \quad \theta \in [0, 1], \quad (7)$$

where $\rho = \pi(1 - \theta)/(1 - \pi\theta)$ is the bequest rate (the log-utility savings rate derived from (4)), and θ is the share of intergenerational transmission that bypasses the tax-and-transfer system. This direct-transmission channel covers the persistent non-financial inheritances — genetic endowments, social networks, professional contacts, family name, cultural capital — that are not income and not taxable. The two-channel form follows the structure of the accumulation equation in [Bénabou](#)

(2000), with θ taking the role of his direct-transmission term. Its substantive role here is to give the inequality dynamic a non-tax-mediated floor, which connects to the wartime destruction of concentrated, hard-to-tax wealth (Section 4.4.4 reports that the war also raises the inheritance top rate, the most direct measure of taxation on intergenerational transfers).

Cross-generation persistence in capital is

$$P(\tau) = \theta + (1 - \theta)\alpha(1 - \tau). \quad (8)$$

Even at $\tau = 1$, $P = \theta > 0$: redistribution does not remove the direct-transmission floor. The cross-sectional variance evolves according to

$$\sigma_{t+1}^2 = P(\tau_t)^2 \sigma_t^2 + (1 - \theta)^2 (1 - \tau_t)^2 v^2, \quad (9)$$

so the steady-state inequality given a fixed τ is

$$\sigma_{ss}^2(\tau) = \frac{(1 - \theta)^2 (1 - \tau)^2 v^2}{1 - P(\tau)^2}, \quad (10)$$

which is strictly decreasing in τ over $[0, 1]$ with $\sigma_{ss}^2(1) = 0$.

Political equilibrium is set by myopic voting: agents vote on τ_t before z_t^i realises and do not internalise the effect of τ_t on future inequality. The pivotal voter is wealth-biased,

$$\ln k_t^{\text{piv}} = m_t + \lambda \sigma_t, \quad (11)$$

where $m_t = E[\ln k_t^i]$, $\sigma_t = \sqrt{\sigma_t^2}$, and $\lambda > 0$ captures the disproportionate political influence of wealthier voters. The pivotal voter's preferred tax rate solves the first-order condition

$$(1 - \tau)(\alpha^2 \sigma^2 + v^2) - \alpha \lambda \sigma - \frac{\delta}{\eta} \cdot \frac{\tau}{1 - \tau} = 0, \quad (12)$$

which implicitly defines the political curve $\tau = T(\sigma^2)$. The three terms have a clean reading: $(1 - \tau)(\alpha^2 \sigma^2 + v^2)$ is the marginal benefit of redistribution, combining an insurance motive proportional to v^2 and an inequality-reduction motive proportional to $\alpha^2 \sigma^2$; $\alpha \lambda \sigma$ is the marginal opposition from the wealth-biased pivotal voter; $(\delta/\eta) \cdot \tau/(1 - \tau)$ is the marginal deadweight loss from the labour distortion. Derivations of the bequest rate, the labour first-order condition, the variance law of

motion, and the political first-order condition are in [Appendix C](#).

5.3 Multiple equilibria

A steady-state political-economic equilibrium is a pair (σ_{ss}^2, τ^*) that satisfies both $\tau^* = T(\sigma_{ss}^2)$ and $\sigma_{ss}^2 = \sigma_{ss}^2(\tau^*)$ — the pivotal voter’s preferred tax rate at the inequality level, and the inequality level sustained by the law of motion under that tax rate. Geometrically, an equilibrium is an intersection of the political curve and the inequality locus in (σ^2, τ) space.

Under the illustrative calibration $\theta = 0.7$, $\alpha = 0.95$, $v^2 = 1.0$, $\lambda = 2.0$, $\delta/\eta = 0.01$ the two curves intersect at three points ([Figure 6](#)): a high-inequality, low-redistribution steady state at $(\sigma^2, \tau) \approx (1.82, 0.03)$; an unstable threshold at $(0.068, 0.524)$; and a low-inequality, high-redistribution steady state at $(0.014, 0.746)$. The two outer steady states are stable in the sense that small perturbations decay back toward them; the middle one is unstable and separates the basins of attraction of the two stable equilibria. The multiplicity is presented here as a property of the calibrated model that gives the war shock its structural foothold; the exact parametrisation and the verified intersection values are in [Appendix C](#).

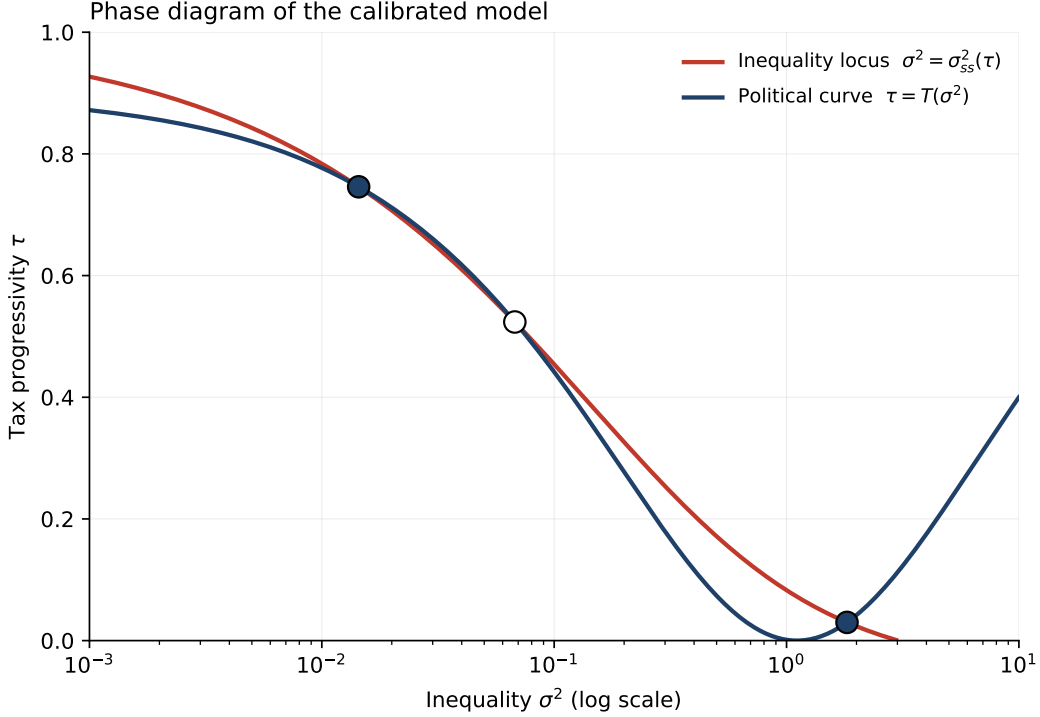


Figure 6: Phase diagram of the calibrated model. The inequality locus $\sigma^2 = \sigma_{ss}^2(\tau)$ (red) and the political curve $\tau = T(\sigma^2)$ (blue) intersect at three points: two stable (filled), and one unstable (open) that separates their basins of attraction. Inequality on a log scale. Calibration $\theta = 0.7$, $\alpha = 0.95$, $v^2 = 1.0$, $\lambda = 2.0$, $\delta/\eta = 0.01$.

5.4 The war as a shock that selects an equilibrium

The war is modelled as a one-time exogenous shock to σ^2 , the cross-sectional variance of log capital. The primitive parameters $(\theta, \alpha, v^2, \lambda, \delta/\eta)$ are unchanged: preferences, the production technology, the wealth-bias mapping from inequality to political influence, and the labour distortion are all unaffected. What changes is the state variable: progressive capital destruction during the war — capital levies, factory destruction, wartime expropriation, the wartime collapse in financial-asset valuations — falls disproportionately on top wealth-holders and compresses the cross-sectional distribution.

Whether the shock has a permanent effect depends on whether it pushes σ^2 below the unstable threshold. If it does not, the economy stays in the basin of the pre-war steady state and inequality and the tax rate drift back to their pre-war values. If it does, the economy enters the basin of the post-war steady state: the political response to the lower σ^2 is an immediate jump in τ , after which the joint dynamics carry σ^2 further down and τ further up toward the post-war steady state.

Figure 7 traces a simulated path for a shock that crosses the threshold: τ jumps from the pre-war value of about 0.03 to about 0.68 in the period of entry, then drifts slowly toward the post-war steady state at about 0.75.

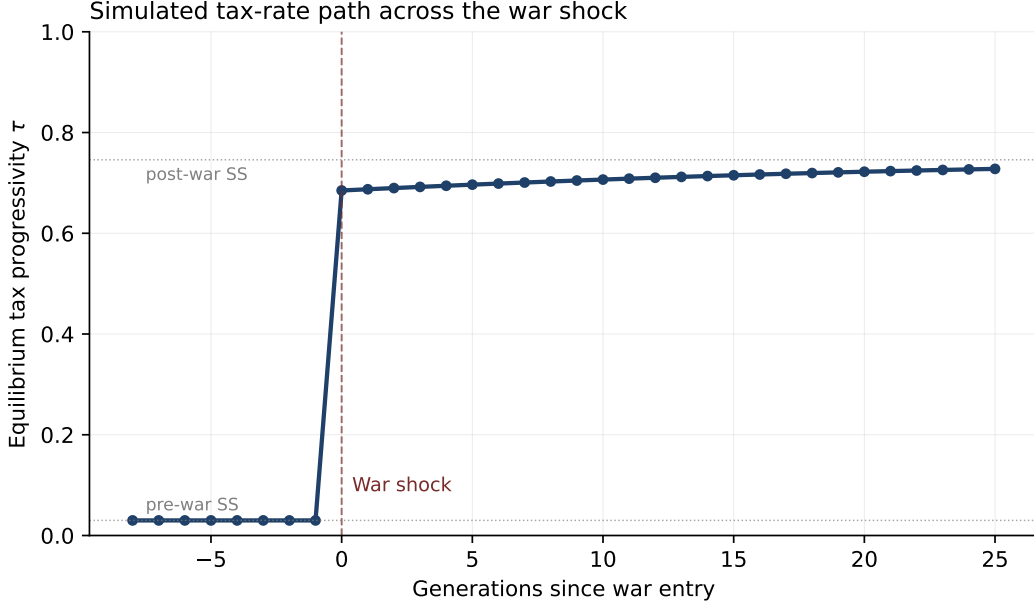


Figure 7: Simulated equilibrium tax-rate path across the war shock. The pre-war steady state $\tau \approx 0.03$ is sustained for several generations; at $t = 0$ a one-time shock displaces σ^2 below the unstable threshold. The political response is immediate (τ jumps to about 0.68), and the joint dynamics carry the economy slowly toward the post-war steady state at $\tau \approx 0.75$.

The contrast with the preference-based account of [Scheve and Stasavage \(2010\)](#) is precise. In their account, the war changed preferences: the experience of mass conscription raised the demand for fairness, and the war-active electorate that emerged after 1945 was different from the one that entered the war. In the account formalised here, preferences are fixed and the war changed the distribution; the unchanged political process, facing a more compressed distribution, sustains a higher level of redistribution, which then sustains the distribution. Persistence is a property of the new equilibrium rather than of altered tastes. The two accounts are distinct and complementary — distributional versus preference-based — and which fits a particular country or period better is for the reader to weigh.

The threshold-crossing structure maps qualitatively to the exposure-category ordering in Section 4.4.5. Larger shocks are likelier to cross the threshold and select the high-redistribution basin; smaller shocks are more likely to leave the economy in the pre-war basin. Empirically, active combatant countries received the largest distributional shocks and show the most durable post-event

divergence, while marginal participants show smaller and less durable effects. The mapping is qualitative — the model does not deliver an exact threshold in observable shock magnitudes — and it is consistent with the differential-persistence pattern documented in [Section 4.4.5](#).

6 Conclusion

The Second World War raised top marginal tax rates sharply and durably in the countries that mobilized for it. Five years after entry, a war-active country's top rate was about 38 percentage points higher than the path implied by the non-war-active control group and the country's own pre-event dynamics; three decades out the effect was about 42 percentage points; the effect remained large and statistically distinct from zero for five decades before fading. The estimates survive a wide range of inference methods and specifications, and the same design applied to the First World War yields a similar but smaller effect. The wartime rise and its multi-decade persistence are the central empirical findings of this paper.

We read the persistence as a shift between political-economic equilibria rather than as a change in preferences. By compressing the wealth distribution at the top, the war moved these countries into a basin of attraction with higher redistribution that then sustained itself: the unchanged political process, facing a more compressed distribution, kept top taxation high, and the high top taxation kept the distribution compressed. The mechanism is distinct from, and complementary to, the fairness-preference account in which the war raised the demand for fairness directly; the paper does not adjudicate between the two, and which fits a given country or period is left for the reader and for further work.

The paper contributes to three literatures. To the literature on warfare and taxation, it traces not only the wartime rise but the full multi-decade trajectory of the effect, including its eventual fade after 1980. To the literature on long-run institutional persistence, it documents that a single transitory shock reshaped a core fiscal institution for decades, with the First World War's similar but smaller effect a direction-consistent second instance. To the literature on cross-country differences in tax progressivity, much of the dispersion that later work treats as given is, in this account, the lasting effect of a single historical shock rather than a settled reflection of present-day preferences or institutions.

Three limitations bound the exercise. The long-horizon estimates are less precise than the short- and mid-horizon ones, because the sample is small (35 countries, 13 of them war-active) and several inference methods designed for large panels lose precision there; the long-horizon point estimates remain of the right sign and order, but the wider confidence intervals at $\tau = 50$ and $\tau = 70$ follow from the limited sample size. The model is an illustration of the equilibrium-shift

mechanism rather than an estimated structural model, and its τ is a progressivity parameter while the empirical object is the top marginal rate; the two co-move, but a fully consistent multi-bracket schedule is left for future work. The post-1980 decline in top rates is documented and is consistent with the account in Section 2 of a broad policy turn and capital-mobility shift, but its drivers are outside the paper’s identification strategy.

The natural next steps follow from those limitations. A model that specifies a progressive schedule and maps to the top marginal rate directly would tighten the connection between the theory and the empirical object. Tracing the interaction between the elevated top rates the war produced and the post-1980 reversal — whether the latter was a fundamental shift in the parameters that govern the post-war equilibrium or a separate downward force operating alongside it — is a question this paper raises but does not settle. The broader point of the exercise is that the progressivity of modern tax systems is, to a substantial degree, a lasting effect of mid-century warfare rather than a fixed feature of present-day politics or institutions; what holds top rates where they are now was in large part produced by a shock eight decades old.

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A Source documentation

Table 8: Top marginal rate source by country

Country	Source for top-rate series	Group
Australia	Comparative Income Taxation Database (CITD)	T
Austria	CITD	C
Belgium	CITD	C
Canada	CITD	T
Denmark	CITD	C
Finland	CITD	T
France	CITD	T
Germany	CITD	T
Ireland	CITD	T
Italy	CITD	T
Japan	CITD	T
Netherlands	CITD	C
New Zealand	CITD	T
Norway	CITD	C
South Korea	CITD (1949–2010); pre-independence (1934–1944) extension from colonial-era Joseon Income Tax Order schedules per Kim and Kim (2014) and Nakaoka (2010)	C
Spain	CITD	C
Sweden	CITD	T
Switzerland	CITD	C
United Kingdom	CITD	T
United States	CITD	T
Argentina	Argentine primary tax statutes (Decreto-Ley 11.586/1932; Ley 11.682/1932 onward) via Infoleg; Alvaredo (2010); Della Paolera and Taylor (2003)	C
Bolivia	Bolivian primary tax law (Ley de 3 mayo 1928 onward) via LexiVox; Drake (1989)	C
Brazil	Brazilian primary tax law (Lei 4.625/1922 onward) via Planalto and Receita Federal Memória; Souza and Medeiros (2015)	C
Chile	Chilean primary tax law (Ley 3996/1924 and Const. 1925 onward) via BCN LeyChile; Biehl, Labarca and Atria (2025); Díaz, Lüders and Wagner (2016)	C
Colombia	Colombian primary tax law (Ley 56/1918 onward) via Diario Oficial; Junguito and Rincón (2004); Alvaredo and Londono (2013)	C
Ecuador	Ecuadorian primary tax law (Ley 1925 onward) via Registro Oficial; Paz y Miño (2015); SRI historical archive	C
El Salvador	Salvadoran primary tax law (Ley de Impuesto sobre la Renta 1915 onward); REDICCES tax-history monograph	C
Iceland	Icelandic primary tax law (Lög nr. 74/1921 onward) via Stjórnartíðindi	C
India	Indian Finance Acts and Income Tax Act archives via the Income Tax Department of India; Banerjee and Piketty (2005); Chancel and Piketty (2019)	C
Indonesia	Dutch East Indies primary statutes (Staatsblad 1908 No. 298 onward); Booth (1998); Lindblad fiscal-history papers	C
Mexico	Mexican primary tax law (Ley del Impuesto sobre la Renta 1925 onward) via Diario Oficial de la Federación; Aboites Aguilar (2003); Unda Gutiérrez (2015); INEGI <i>Estadísticas Históricas de México</i> (2014)	C
Peru	Peruvian primary tax law (Ley 7904/1934 onward) via SUNAT historical archive	C
Philippines	Philippine primary tax law (Act 1189/1904 onward) via the Bureau of Internal Revenue and Official Gazette	C
Portugal	Portuguese primary tax law (Imposto Pessoal de Rendimento 1922 onward) via Diário do Governo; Alvaredo (2010); Valério (2001)	C
South Africa	South African Income Tax Acts (Act No. 28 of 1914 onward); Alvaredo and Atkinson (2014); Hattingh (2017)	T

Notes: CITD denotes the Comparative Income Taxation Database (Genovese et al., 2016). “Group” classifies each country as war-active (T) or non-war-active (C) for the analysis; the classification is set out in Section 3.2.

B Empirical analysis: additional tables

Table 9: Leave-one-country-out jackknife: full table

Dropped	Treated?	$\rho+\delta$	Δ_5	Δ_{10}	Δ_{15}	Δ_{20}	Δ_{30}	Δ_{30} shift
ITA	T	0.902	33.25	34.03	35.16	36.00	36.19	-6.13
AUS	T	0.898	33.67	38.00	36.69	38.00	39.81	-2.51
JPN	T	0.897	37.48	40.65	38.70	39.28	40.66	-1.66
IND	C	0.892	37.02	39.32	38.52	39.51	41.16	-1.16
CAN	T	0.894	36.52	39.66	38.68	39.67	41.26	-1.06
NZL	T	0.896	36.54	38.98	38.75	39.76	41.50	-0.82
PRT	C	0.892	37.24	39.64	38.90	39.93	41.61	-0.71
PER	C	0.892	37.37	39.83	39.13	40.17	41.87	-0.45
MEX	C	0.892	37.39	39.86	39.17	40.21	41.91	-0.41
BRA	C	0.892	37.43	39.92	39.23	40.29	41.99	-0.33
NOR	C	0.892	37.52	40.04	39.37	40.44	42.16	-0.17
SLV	C	0.892	37.52	40.04	39.38	40.44	42.16	-0.16
ECU	C	0.892	37.56	40.10	39.44	40.51	42.24	-0.09
IDN	C	0.892	37.59	40.14	39.49	40.57	42.30	-0.03
BOL	C	0.892	37.59	40.15	39.51	40.59	42.31	-0.01
ARG	C	0.892	37.62	40.20	39.56	40.64	42.37	+0.05
CHE	C	0.892	37.63	40.21	39.57	40.66	42.39	+0.07
DNK	C	0.893	37.65	40.23	39.60	40.68	42.42	+0.10
PHL	C	0.893	37.67	40.26	39.63	40.72	42.46	+0.13
COL	C	0.893	37.67	40.27	39.64	40.73	42.47	+0.15
ZAF	T	0.889	36.93	40.33	39.41	40.42	42.48	+0.15
AUT	C	0.893	37.71	40.32	39.70	40.80	42.54	+0.22
ISL	C	0.893	37.76	40.39	39.78	40.89	42.64	+0.31
BEL	C	0.893	37.76	40.40	39.80	40.90	42.65	+0.33
ESP	C	0.893	37.78	40.42	39.82	40.93	42.68	+0.36
KOR	C	0.893	37.82	40.49	39.90	41.01	42.77	+0.45
FIN	T	0.889	39.94	40.05	40.78	41.90	43.06	+0.73
CHL	C	0.893	38.00	40.74	40.20	41.34	43.12	+0.80
IRL	T	0.894	38.86	41.41	40.13	41.48	43.18	+0.86
FRA	T	0.889	38.00	41.50	40.63	41.67	43.35	+1.03
NLD	C	0.893	38.18	41.01	40.52	41.68	43.49	+1.17
SWE	T	0.892	38.04	40.72	40.46	41.63	43.50	+1.18
USA	T	0.889	39.04	40.72	39.55	40.46	43.90	+1.58
DEU	T	0.885	40.42	42.67	41.63	42.52	44.68	+2.36
GBR	T	0.887	39.59	42.48	41.96	43.60	45.19	+2.87

Notes: Re-estimate the main specification dropping each of the 35 countries in turn. Sorted by Δ_{30} shift relative to the full-sample main estimate (+42.32). LOO median +42.42; LOO range [+36.19, +45.19].

Table 10: Acemoglu–Naidu–Restrepo–Robinson long-run multiplier (within-OLS, 35-country sample)

Lag order p	$\hat{\beta}$ (pp)	$\hat{\rho}_{\text{sum}}$	$\hat{\beta}_{LR}$ (pp)	95% CI (OLS iid)	95% CI (CR1)
1	1.79	0.941	30.23	[+17.35, +43.10]	[+12.28, +48.18]
2	1.73	0.935	26.55	[+15.04, +38.06]	[+11.36, +41.74]
3	1.74	0.933	25.77	[+14.55, +37.00]	[+11.43, +40.12]
4	1.75	0.932	25.64	[+14.47, +36.81]	[+11.48, +39.80]

Notes: Estimating equation $y_{i,t} = \beta \text{Post}_{i,t} + \sum_{j=1}^p \gamma_j y_{i,t-j} + \alpha_i + f(t) + \varepsilon_{i,t}$ with $f(t)$ a linear–quadratic trend;

$\beta_{LR} = \beta / (1 - \sum_j \gamma_j)$, with confidence intervals by the delta method.

Table 11: Borderline classification: full Δ_τ table with 95% CIs

	Δ_0	Δ_5	Δ_{10}	Δ_{15}	Δ_{20}	Δ_{30}	Δ_{40}	Δ_{50}	Δ_{70}
(a) Strict (drop SWE, IRL); $n_T = 11$; $\rho + \delta = 0.894$; pre- F $p = 0.83$									
point	+9.20	+39.90	+42.77	+41.97	+43.51	+45.43	+38.86	+22.90	+17.24
CI lo	+5.31	+25.42	+26.82	+27.01	+27.92	+27.60	+18.78	+1.54	-2.65
CI hi	+13.07	+53.52	+55.13	+51.98	+54.01	+57.62	+52.13	+37.82	+30.29
(b) Inclusive (add AUT); $n_T = 14$; $\rho + \delta = 0.902$; pre- F $p = 0.90$									
point	+8.98	+35.17	+37.85	+38.33	+38.61	+41.00	+34.77	+20.00	+13.30
CI lo	+5.39	+21.19	+22.58	+23.91	+22.29	+22.29	+15.00	-0.26	-6.45
CI hi	+12.67	+48.49	+50.22	+48.35	+49.55	+53.51	+48.99	+34.84	+28.32
(c) Combined (drop SWE/IRL, add AUT); $n_T = 12$; $\rho + \delta = 0.902$; pre- F $p = 0.62$									
point	+8.55	+37.36	+40.55	+41.20	+41.81	+44.54	+39.57	+24.03	+18.90
CI lo	+5.02	+22.27	+24.10	+26.46	+25.65	+25.02	+18.92	+2.16	-2.10
CI hi	+12.30	+50.86	+53.30	+51.21	+52.57	+56.80	+53.60	+39.89	+33.07

Notes: Three classification variants of the main 35-country specification (sample fixed at 35 countries; only the treated/control split varies). Wild cluster bootstrap 95% confidence intervals (999 replications, Rademacher weights).

B.1 Heterogeneity-robust estimator for staggered timing

When entry is staggered across cohorts, pooled event-study coefficients can be contaminated by comparisons that use already-treated units as controls. The interaction-weighted estimator of [Sun and Abraham \(2021\)](#) avoids this by forming the effect for each entry cohort separately and then averaging, so no already-treated cohort serves as a control for a later one. The resulting per-period path is almost identical to the pooled one — a mean absolute gap of 0.31 percentage points (Table 12) — so heterogeneous entry timing is not distorting the estimates.

Table 12: Sun–Abraham heterogeneity-robust per-period coefficients vs. pooled event study

τ	Pooled (TWFE)	Sun–Abraham	Absolute gap
−5	+0.50	−0.74	1.24
−4	+2.50	+1.51	0.99
−3	+2.09	+2.10	0.02
−2	+0.76	+1.36	0.60
0	+6.86	+6.89	0.03
5	+16.18	+15.41	0.77
10	+12.39	+12.85	0.46
15	+5.42	+5.00	0.42
20	+4.15	+3.84	0.31
30	−1.64	−1.55	0.09
40	−9.44	−9.87	0.43
50	−14.01	−15.74	1.73

Notes: Per-period β_τ under the main specification’s pooled event study (two-way fixed effects) and the heterogeneity-robust interaction-weighted estimator of [Sun and Abraham \(2021\)](#).

B.2 Sensitivity to parallel-trends violations

A pre-trend test examines only the pre-period. To gauge how far a post-period violation of parallel trends would change the conclusions, we apply the HonestDiD analysis of [Rambachan and Roth \(2023\)](#), which asks how large such a violation would have to be before a per-period coefficient ceased to be distinct from zero (Table 13). The breakdown value $\bar{M}$ records that threshold relative to the largest violation seen in the pre-period: $\bar{M} = 1$ means a coefficient survives a violation as large as the worst pre-period one, and larger values mean greater robustness. The entry-year effect is the most robust ($\bar{M} = 1.2$), and the mid-horizon coefficients withstand violations of roughly 40 to 50 percent of the worst pre-period deviation. The effect therefore does not require parallel trends to hold exactly: it survives departures of the size the pre-period suggests is plausible, and because the cumulative effect carries the entry-year impulse forward through persistence, it does not depend on any single per-period coefficient.

Table 13: HonestDiD per-period breakdown values

τ	$\hat{\beta}_\tau$ (pp)	CR1 95% CI	Breakdown $\bar{M}_{RM}$
0	+9.66	[+5.60, +13.72]	1.2
5	+6.75	[-1.50, +14.99]	0.0
10	+2.40	[-2.53, +7.32]	0.0
15	+3.65	[+1.26, +6.04]	0.4
20	+4.13	[+1.33, +6.93]	0.4
30	+4.63	[+1.75, +7.51]	0.5

Notes: Per-period $\hat{\beta}_\tau$ from the main specification with its CR1 95% confidence interval, and the relative-magnitudes breakdown value $\bar{M}_{RM}$ of [Rambachan and Roth \(2023\)](#) — the largest pre-period-relative violation of parallel trends at which the robust 95% confidence set for β_τ still excludes zero.

B.3 Effect by wartime-exposure category

The two-way war-active/non-war-active split used in the main analysis groups together countries with very different wartime experiences. To see whether the rise in top rates was confined to active combatants, we divide wartime exposure into three categories — active combatants (the thirteen war-active countries, which mobilized and fought), invaded non-combatants (countries occupied without mobilizing their own populations), and non-active participants (countries that contributed below the mobilization threshold or stayed neutral, including neutral countries such as Switzerland that mobilized without engaging in combat) — and compare the average top rate before the war (1929–1938) with the early postwar average (1945–1955) in each (Table 14). All three groups raised top rates, but by markedly different amounts: active combatants by about 29 percentage points, invaded non-combatants by about 21, and non-active participants by about 18. The ordering — largest for the most heavily mobilized — is what the mechanism predicts, and the rise is clearly not confined to active combatants. These are raw pre-versus-post differences rather than the identified event-study effect: they describe the pattern across exposure levels, and the smaller, heterogeneous categories do not support a separate causal estimate for each.

Table 14: Average top marginal rate before and after the war, by wartime-exposure category

Exposure category	Countries	Pre-war mean	Postwar mean	Change
Active combatant	13	42.4	71.0	+28.6
Invaded non-combatant	9	31.8	53.0	+21.2
Non-active participant	13	13.8	31.3	+17.6

Notes: Unweighted cross-country mean of the top marginal income tax rate (percentage points) in the pre-war period (1929–1938) and the early postwar period (1945–1955), and the change between them, for the three wartime-exposure categories. These are raw group means, not regression-adjusted estimates.

B.4 Common parametric trends

A stricter way to capture the slow movement of top rates is to replace the lagged dependent variable with a parametric trend in calendar time, absorbing that movement directly rather than through persistence. Dropping the lagged rate and its interaction with the post-entry indicator from equation (1) and adding a common quadratic or linear trend — a single shared path across countries, with the country fixed effects retained — gives the quadratic-trend specification

$$\text{toprate}_{i,t} = \sum_{\tau \in \mathcal{T}} \beta_{\tau} \cdot \mathbf{1}[e_{i,t} = \tau] + \alpha_i + \gamma_1 \cdot t + \gamma_2 \cdot t^2 + \varepsilon_{i,t}, \quad (13)$$

and the linear-trend specification

$$\text{toprate}_{i,t} = \sum_{\tau \in \mathcal{T}} \beta_{\tau} \cdot \mathbf{1}[e_{i,t} = \tau] + \alpha_i + \gamma_1 \cdot t + \varepsilon_{i,t}, \quad (14)$$

where t is the year (centered for numerical conditioning) and γ_1 and γ_2 are linear and quadratic trend coefficients common to all countries. These specifications report per-period coefficients β_{τ} , not the cumulative Δ_{τ} . The quadratic and linear trends give entry-year effects of $\beta_0 = +6.0$ and $+9.9$ percentage points, against $+9.7$ for the main specification and $+6.9$ under year fixed effects, and all show large positive per-period coefficients through the first decade (Figure 8). They diverge later, for a reason that lies in how the per-period coefficients are constructed rather than in the economics: a per-period coefficient measures the gap between the war-active countries' actual rate and the fitted path (country fixed effects plus the common trend), so once that trend has absorbed the broad post-1980 fall in top rates, the war-active rates — having climbed from a higher base — register as a negative deviation from it at long horizons. The year-fixed-effects and quadratic

per-period paths therefore turn negative beyond $\tau \approx 30$, whereas the cumulative effect of the main specification, which compounds the early effect through persistence, stays elevated; the contrast reflects what each object measures, not a reversal of the war's effect. All three pass the joint pre-trend test, the linear trend least comfortably, since a single straight line cannot fit both the wartime rise and the later fall (joint pre-trend $p = 0.27$, against 0.92 for the quadratic trend and 0.93 for year fixed effects); we accordingly give the linear trend the least weight.

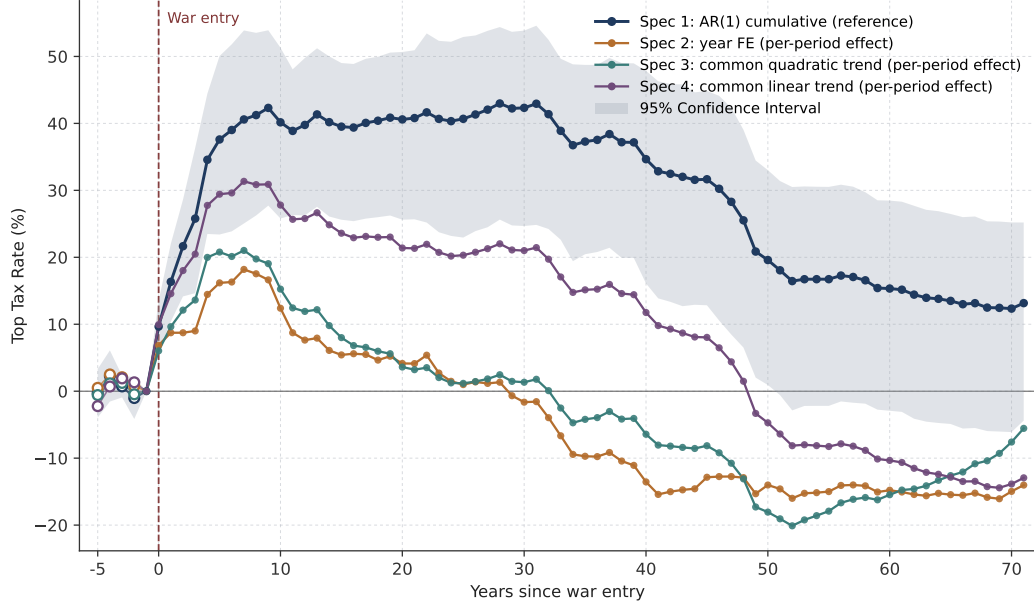


Figure 8: Cumulative effect of the main specification versus per-period coefficients from the trend specifications. The solid line is the cumulative Δ_τ from the main specification; the remaining series are the per-period coefficients β_τ from the year-fixed-effects (equation 3), common-quadratic-trend (equation 13), and common-linear-trend (equation 14) specifications.

C The model: derivations and parametrization

This appendix derives the equations of Section 5 and reports the calibration and verified steady states.

Bequest rate (the derivation of equation 7)

The bequest rate ρ in (7) is the optimal savings rate of an adult with log utility (4). Let s denote the savings rate, so the financial bequest is $e = s\hat{y}$ and consumption is $c = (1 - s)\hat{y}$. Substituting

into (4) and using (7),

$$U_t^i = (1 - \pi) \left[\ln((1 - s)\hat{y}_t^i) - \frac{(\ell_t^i)^\eta}{\eta} \right] + \pi [\theta \ln k_t^i + (1 - \theta) \ln(s\hat{y}_t^i)].$$

The first-order condition with respect to s is

$$-\frac{1 - \pi}{1 - s} + \frac{\pi(1 - \theta)}{s} = 0 \quad \Longleftrightarrow \quad s = \frac{\pi(1 - \theta)}{1 - \pi\theta} \equiv \rho.$$

Labour supply (the derivation of $\ell = \chi(1 - \tau)^{1/\eta}$)

Given $s = \rho$, the agent's labour first-order condition with the production function (5) and the tax scheme (6) reduces to

$$\ell_t^i = \chi(1 - \tau_t)^{1/\eta}, \quad \chi = \left[\frac{\delta(1 - \pi\theta)}{1 - \pi} \right]^{1/\eta},$$

so labour falls in the tax rate with Frisch elasticity $1/(\eta - 1)$.

Variance law of motion (the derivation of equation 9)

Take the variance of $\ln k_{t+1}^i$ from (7). Using $\ln \hat{y}_t^i = (1 - \tau_t) \ln y_t^i + \tau_t \ln \tilde{y}_t$, the fact that $\ln \tilde{y}_t$ is common across i , and $\ln y_t^i = \ln z_t^i + \alpha \ln k_t^i + \delta \ln \ell_t^i$ with $\ln \ell_t^i$ common across i ,

$$\text{Var}(\ln y_t^i) = \alpha^2 \sigma_t^2 + v^2, \quad \text{Cov}(\ln k_t^i, \ln \hat{y}_t^i) = \alpha(1 - \tau_t) \sigma_t^2.$$

Substituting,

$$\sigma_{t+1}^2 = \theta^2 \sigma_t^2 + 2\theta(1 - \theta)\alpha(1 - \tau_t) \sigma_t^2 + (1 - \theta)^2(1 - \tau_t)^2(\alpha^2 \sigma_t^2 + v^2),$$

which factors to (9) on collecting σ_t^2 terms.

Political first-order condition (the derivation of equation 12)

The pivotal voter's ex-ante expected utility, taking expectations over z_t^i before voting, is

$$V^{\text{piv}}(\tau; \sigma^2) = \text{const} + (1 - \pi\theta) E_z[\ln \hat{y}^{\text{piv}}] - (1 - \pi) \frac{\chi^\eta (1 - \tau)^\eta}{\eta},$$

where the coefficient $(1 - \pi\theta)$ on $E_z[\ln \hat{y}^{\text{piv}}]$ collects the bequest weight from (7) and the consumption weight from (4) under the optimal ρ . Substituting the after-tax income (6), the labour first-order condition, and $\ln k^{\text{piv}} = m + \lambda\sigma$, and differentiating in τ , gives the political first-order condition (12).

Calibration and verified steady states

The figures and the steady states reported in Section 5 use the illustrative calibration

$$\theta = 0.7, \quad \alpha = 0.95, \quad v^2 = 1.0, \quad \lambda = 2.0, \quad \delta/\eta = 0.01.$$

The calibration is not estimated; it is chosen to place the model in the multiple-equilibria regime where the war-shock argument is operative. Solving (12) for τ at each σ^2 and intersecting with the inequality locus (10) returns three steady states (Table 15); the algebra of equations (8)–(12) reproduces these intersections to four decimal places.

Table 15: Verified steady states of the calibrated model			
	Inequality σ^2	Tax rate τ	Stability
Pre-war steady state	1.821	0.030	stable
Unstable threshold	0.068	0.524	unstable
Post-war steady state	0.014	0.746	stable

Notes: Computed from the verified model at the illustrative calibration. The two stable steady states are local attractors under the joint dynamics of (9) and (12); the unstable threshold separates their basins of attraction.